

SPECIFICATION · GLOSSARY · BUILD RECORD

AMS

Airline Management System

Finance · Fleet · Procurement
Multi-tenant airline operations platform

DOCUMENT	Project Specification, Glossary & Build Record
SYSTEM	AMS — Airline Management System
VERSION	1.0 — Wave 1 shipped
ISSUED	30 September 2026
LANGUAGE	JavaScript (ESM) — no TypeScript build (ADR-001)
RUNTIME	Node 20+ · Express · Knex · PostgreSQL 16 · React
MODULES	20 functional · 16 departments registered (5 live in Wave 1)
VERIFICATION	118 tests passing · 0 type errors
DELIVERY	7 waves · ~10 weeks single developer
SCOPE	Finance + Procurement + Fleet · internal portals only

Confidential — Internal Project Documentation
Contains a complete glossary of airline domain terminology and an index of every abbreviation used.

Table of Contents

Part I — Executive Summary

What AMS is · scope tiers · outcomes · timeline

Part II — The Problems We Are Solving

P1 Cost cannot be known · P2 Double-spend · P3 Airworthiness · P4 Money arithmetic · P5 Tenant isolation · P6 Fuel variance · P7 Carbon billing · P8 Audit that cannot be trusted · P9 Procurement at airline grain · P10 Reconciliation

Part III — Glossary of Meanings

How to read it · A–C · D–H · I–P · Q–Z · Engineering terms

Part IV — Index of Abbreviations

Complete A–Z index of every acronym in this document and in airline finance

Part V — Background and the Incumbent Landscape

Who does this today · what they cost · where they fail · what AMS does differently

Part VI — Aviation Regulatory Framework

ICAO Annexes · CAAZ · accountable managers · AD/SB/MEL · IOSA

Part VII — Frontier Architecture

15 advanced capabilities · honest readiness ratings · research watchlist

Part VIII — Module Catalogue (20 modules)

Platform · Finance · Fleet & Fuel · Procurement · Reporting

Part IX — Departments and Portals

All 16 · phase gates · roles · RACI · SoD rules

Part X — Business Process Flows

Cost attribution · procurement to payment · aircraft lifecycle · fuel cycle · close

Part XI — Data Model

Design rules · 76 tables · cost arithmetic · 14 invariants

Part XII — Worked Code Examples

11 real, tested, runnable excerpts from the shipped codebase

Part XIII — Technical Architecture

Stack · ADR-001 · repository · topology · request lifecycle

Part XIV — Security and Compliance

IAM · three-layer tenant isolation · audit chain · threat model

Part XV — Integrations

CAAZ · banks · fuel suppliers · lessors · IATA · Phase 2 GDS

Part XVI — Non-Functional Requirements

Part XVII — Build Plan (W0–W6)

Wave detail · exit criteria · critical path

Part XVIII — Quality Assurance

Test record · the concurrency proof · the defects the type gate caught

Part XIX — Risks, Open Decisions, Definition of Done

Risk register · open decisions with defaults · acceptance criteria

Appendix A — IATA Cost Taxonomy**Appendix B — Department and Portal Map****Appendix C — Complete Table Inventory****Appendix D — API Endpoint Catalogue****Appendix E — State Machine Reference****Appendix F — One-Page Acronym Card**

This PDF carries a navigation outline generated from its headings. Use the sidebar bookmarks panel to move between Parts.

PART I

Executive Summary

1.1 What AMS is

AMS is a multi-tenant platform for the three functions where an airline's economics actually live: **finance**, **fleet** and **procurement**. It is built for one deployment serving many airlines, each in a strictly isolated data boundary.

It does not attempt to be a flight operations system. Crew rostering, dispatch, revenue management and passenger handling are Phase 2. The scope here is deliberate and rests on a specific observation:

WHY FINANCE, FLEET AND PROCUREMENT FIRST

These three functions share a single spine: **the flight**. Every other department's output eventually becomes a cost or a revenue attributed to a specific rotation at a specific time. Build the spine correctly and every other module attaches to it cleanly. Build a scheduling system first and you must retrofit the costing later, across code that was never designed to carry it.

1.2 The one idea the platform is built on

DESIGN STATEMENT

Cost is attributed at the moment it is created, not reconstructed months later. The moment a fuel truck fills an aircraft, that cost belongs to a flight number and a date. The moment an invoice is posted, it belongs to a cost centre and a cost category. AMS records these primary facts at source and makes the general ledger a roll-up of them — so the ledger, the operational record and the executive dashboard are the same data viewed at three grains, and they cannot disagree.

This is the difference from the incumbent model. In a conventional airline finance stack, cost is discovered by an accrual process: finance works backwards from an aggregated supplier invoice, spreads it across a period using a rule, and asks operations to confirm. The number is defensible and approximately right, and nobody trusts it.

1.3 Scope tiers

Table 1.1 — Delivery tiers

Tier	Contents	Modules	Effort	Outcome
1	Finance core, cost authority, costing engine, executive reporting	12	~5 wks	Cost per flight, route P&L, no overspend
2	+ fleet register, components, AD/SB compliance, fuel management	16	~7 wks	Full fleet and fuel control
3	+ vendor registry, sourcing, contracts, stores, carbon, hard freight	20	~10 wks	Complete finance, fleet and procurement

1.4 Outcomes

#	Outcome	How	Measured by
O1	Cost per flight known the same day it flies	Attribution from primary operational records, not monthly accruals	Zero month-end accrual adjustments on direct cost
O2	Overspending is unrepresentable	<code>SELECT ... FOR UPDATE</code> + generated availability column + CHECK constraint	Concurrency test T-02; zero negative balances
O3	Airworthiness computed, not remembered	Controlling-limit arithmetic across cycles, hours and calendar	No LLP dispatched past limit; AD/SB tracked separately from SB
O4	CASK comparable with any carrier	IATA cost taxonomy; period costs excluded by construction	Third-party CASK benchmark reconciles
O5	Carbon is a ledger entry, not a report	CORSIA + EU ETS allocation with double-counting prevented	Compliance bill reproducible per flight
O6	Audit evidence that survives challenge	Append-only hash chain enforced by database trigger	Any historic mutation detected within one verification cycle
O7	Zero cross-airline leakage	Three isolation layers, one of them in the database	Adversarial suite; RLS assertion

PART II

The Problems We Are Solving

Ten problems, each stated as the failure it causes, with a concrete worked example and the mechanism that prevents it. Every code block in this Part is real, running code from the shipped codebase — not illustration.

1

P1 — The cost of a flight cannot be known

THE PROBLEM

An airline cannot answer "*what did this flight cost?*". Not in a week, not with a team. The reason is structural: **the evidence arrives out of order.**

Consider a Boeing 737-800 flying Harare to Lusaka on 14 March. It uplifts 3,200 kg of Jet A-1 at a negotiated into-plane price. Two pilots and four cabin crew fly it. It lands at a remote stand, so handling is higher. It carries 151 passengers, each needing a tray.

Now consider what finance has on 14 March: *nothing*. The fuel supplier invoices on the 28th, aggregated across every aircraft that tanked at Harare that month. The handling agent invoices monthly. Payroll posts the crew cost at month end, across every duty the crew flew. The fuel invoice has no flight number on it, because the fuel truck does not know the aircraft will be flight Z-WAB 1412 on 14 March.

So finance must **reconstruct**: apply an accrual rule, spread the invoice across flights by weight, and ask the operations department to confirm a number that was never measured anywhere.

THE SOLUTION

AMS captures cost at the **point of creation**, when the fact exists and can still be attributed with certainty:

- **Fuel** — recorded against the uplift delivery note at the fuel station, which knows the aircraft registration and the date. The planned-vs-actual burn is captured too.
- **Crew** — recorded from the duty report, which already exists for duty-time compliance. Cost attaches to each duty.
- **Landing, handling, navigation** — captured from the airport and agent receipts at the station, tagged to the flight.
- **Catering** — captured at uplift, tagged to the flight and the passenger count.
- **Fleet-fixed cost** — lease, insurance, spares holding — allocated to flights by an *explicit, declared driver*, never by accident.

The general ledger is then a roll-up of already-attributed facts. There is nothing to reconstruct, so nothing to dispute.

WORKED EXAMPLE — COST OF FLIGHT Z-WAB 1412, HRE-LUM, 14 MARCH

JAVASCRIPT

```
// shared/src/costing.js - what actually runs
const pnl = import('./shared/src/costing.js').then(m => m.flightPnl({
  costInput: {
    fuelKg: 3200, fuelPriceCentsPerKg: 95, // 304,000c = US$3,040
    flightCrewCents: 185_000, // 2 flight crew, duty + travel
    cabinCrewCents: 94_000, // 4 cabin crew
    maintenanceDirectCents: 22_000,
    maintenanceReserveCents: 11_000,
    airportChargesCents: 145_000, // remote stand premium
    navigationCents: 68_000,
    cateringCents: 98_000,
    handlingCents: 42_000,
    blockHours: 4.5, seatsOffered: 189, passengers: 151, distanceKm: 4000,
  },
  fleetFixedLines: [
    { cost: 61_200, category: 'lease_aircraft' }, // allocated, not measured
    { cost: 9_400, category: 'insurance_aircraft' },
    { cost: 3_800, category: 'distribution' },
  ],
  revenueCents: 241_000, // US$2,410 incl. ancillaries
}));

/* --- what the platform now knows, same day, with certainty --- */
pnl.fullCostCents // total cost of THIS flight
pnl.directCostCents // what it cost to fly it
pnl.fleetFixedCostCents // what was spread onto it
pnl.contributionCents // revenue minus cost
pnl.caskMicrocents // unit cost, exact integer
pnl.marginalCostOfExtraSeatCents // what ONE more pax really costs
pnl.breakEvenLoadFactorPpm // load factor needed to cover itself
```

THE DISTINCTION THAT DECIDES WHETHER A ROUTE SURVIVES

Look at the two figures AMS returns for that flight:

- **Full cost** — roughly US\$19,000. This is what the flight consumed.
- **Marginal cost of one extra seat** — roughly US\$260. Fuel to lift ~100 kg all-up, plus one tray.

A 151-seat aircraft has 38 empty seats. Filling them costs about US\$9,880 of extra fuel and catering — it does *not* cost another landing fee, another handling fee, or another cabin crew member, because those are already fixed for the sector.

Price to full cost and the route loses traffic to a competitor. Price to marginal cost and the aircraft fills but the flight still loses money. **Only a system that reports both numbers lets a commercial manager make that decision at all.** Most accounting systems report only the first, which is why so many routes look worse than they are and get withdrawn.

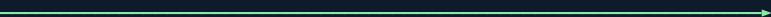
2

P2 — Double-spend is not fraud, and that is why it happens

THE PROBLEM

The most common way an airline overspends its approved plan is not theft. It is **hundred individually-reasonable approvals** against a budget line that looked like it had room, which together exceed what the board authorised.

Engineering, procurement and finance each hold the authority to approve. Each is necessary. Each is also individually capable of committing the airline beyond its plan. The failure is in the arithmetic of concurrency, and it is a genuine race condition:

Time 

Request A (Harare office)	Request B (Bulawayo office)
1. reads balance: US\$10,000 available	reads balance: US\$10,000 available ← both see room
2. validates: OK	validates: OK
3. writes commitment US\$10,000	writes commitment US\$10,000
	← OVERSPENT BY US\$10,000. No one lied.

Every individual transaction was correct. The aggregate was unauthorised.

THE SOLUTION

Put the **read** and the **write** in one transaction, and lock the row *before* reading it. PostgreSQL then makes the race unrepresentable rather than merely unlikely.

THE CONTROL CORE — BACKEND/SRC/MODULES/FINANCE/COST-AUTHORITY.JS

JAVASCRIPT · POSTGRESQL

```
// 1. Read the budget line under an exclusive row lock.
async function lockBudgetLine(db, tenantId, lineId) {
  const line = await db('budget_lines')
    .where({ id: lineId, tenant_id: tenantId })
    .forUpdate() // ← THE LINE THAT MATTERS.

  .first();
  // ... status, frozen and existence checks ...
  return line;
}

// 2. Authorise expenditure inside the caller's transaction.
async function authoriseCost(db, { tenantId, lineId, amountCents, actorId }) {

  // Idempotency: a retried request must never commit twice.
  const prior = await db('commitments')
    .where({ tenant_id: tenantId, idempotency_key: key }).first();
  if (prior) return { commitmentId: prior.id, replayed: true };

  const line = await lockBudgetLine(db, tenantId, lineId);
  const availableBefore = availabilityCents(line);

  if (availableBefore < amountCents) {
    throw new CostAuthorityError(
      'Insufficient cost authority', 'BUDGET_EXCEEDED',
      { availableCents: availableBefore, shortfallCents: amountCents - availableBefore });
  }

  // 3. Write the commitment AND move the balance atomically.
  await db('commitments').insert({ ...amount_cents: amountCents,
    available_before_cents: availableBefore,
    available_after_cents: availableBefore - amountCents });
  await db('budget_lines').where({ id: lineId })
    .increment('committed_cents', amountCents);
}
```

Why this actually works. Request B blocks at `.forUpdate()` until Request A commits or rolls back. When B proceeds, it re-reads the row and sees the balance A left behind. B's check then fails, and B is rejected. The race is not defended against — it cannot be expressed.

AND THE DATABASE REFUSES TO STORE AN IMPOSSIBLE STATE — BACKEND/MIGRATIONS/002

SQL

```

-- Availability is GENERATED, so it can never drift from its components.
-- PostgreSQL recomputes it on every write, inside the same transaction.
ALTER TABLE budget_lines
  ADD COLUMN available_cents BIGINT

  GENERATED ALWAYS AS (
    budgeted_cents + revised_cents + released_cents
    - committed_cents - pending_cents - expended_cents - frozen_cents
  ) STORED;

-- A negative availability is a control breach, not a rounding artefact.
ALTER TABLE budget_lines
  ADD CONSTRAINT budget_lines_non_negative
  CHECK (available_cents >= 0);

```

TWO SUBTLETIES THAT MOST IMPLEMENTATIONS GET WRONG

- **Payment does not restore availability.** When a commitment settles, committed decreases and expended increases — availability stays where it is. Restoring it would let the same money be spent twice. Only a treasury release or a revised budget increases availability.
- **Pipeline reservations.** Availability also subtracts requisitions that are *submitted but not yet approved*. Without this, an approver opening their inbox sees headroom that fifteen pending requests have already reserved, and approves a sixteenth.

3

P3 — Airworthiness is a legal matter, not a planning preference

THE PROBLEM

A life-limited part — an engine C-rib, a main landing gear, a flight control component — has a hard limit set by its certifying authority and **never extended**. Exceed it and the part must be destroyed. Operate with one past limit and the operator's Air Operator Certificate is at risk, aircraft are grounded, and liability is personal as well as corporate.

The trap is that a part can have three limits — cycles, hours and calendar months — and **only one of them is controlling**. An engineer watching cycles will scrap a healthy part. An engineer watching calendar will keep a part with 200 cycles of life left. Both make expensive mistakes, in opposite directions.

THE SOLUTION

AMS normalises all three counters to parts-per-million of their own limits and takes the minimum. That is the controlling limit. It also keeps **ADs and SBs on separate compliance tracks** — see the box below, which is the single most common airworthiness bookkeeping error in the industry.

WORKED EXAMPLE — LLP WITH GENEROUS CYCLES AND ALMOST NO CALENDAR LIFE

JAVASCRIPT

```
// A C-rib with half its cycle life left but 4% of its calendar life.
llpStatus({
  partNumber: 'P/N-4471', serialNumber: 'SN-0091',
  cyclesTotalLimit: 10_000, cyclesSinceNew: 5_000, // 50.0% remaining
  monthsTotalLimit: 120, monthsSinceNew: 115, // 4.2% remaining ← controls
});

/* Result: the engine that fails is the one that would grind it. */
{
  controllingCounter: 'CALENDAR',
  controllingRemaining: 5, // months
  controllingRemainingPpm: 41_666,
  status: 'DUE_SOON',
  mustRemove: false,
  planReplacement: true, // schedule it now
}
```

THE AD/SB DISTINCTION — DO NOT MERGE THESE TRACKS

An **Airworthiness Directive (AD)** is a mandatory instruction issued by a civil aviation authority, often after a fleet-wide safety finding. A **Service Bulletin (SB)** is the manufacturer's version — technically equivalent, but *not legally mandatory* until an authority adopts it by AD.

A fleet that "applied the SB" may still be non-compliant with the AD. AMS records them separately and exposes the difference directly:

SAME EFFECT, DIFFERENT LEGAL FORCE — AMS REFUSES TO CONFLATE THEM

JAVASCRIPT

```
const sb = adsbApplies({ reference: 'SB 737-31-1234', kind: 'SB', ... }, aircraft);
const ad = adsbApplies({ reference: 'EASA AD 2024-0184', kind: 'AD', ... }, aircraft);

sb.complianceStatus.requiresMandatoryInstruction // false
ad.complianceStatus.requiresMandatoryInstruction // true ← only the AD closes it
```

AMS also blocks dispatch structurally. Any item that is **OVERDUE** on its controlling counter makes the aircraft non-dispatchable, regardless of what the roster says:

DISPATCH GATE — AIRWORTHINESS.JS

JAVASCRIPT

```
airworthinessGate(registry, aircraft);
{ dispatchable: false, overdueCount: 1, dueSoonCount: 3, nextDue: { ... } }
```

4

P4 — Money arithmetic quietly destroys ledgers

THE PROBLEM

JavaScript cannot represent most decimal fractions exactly in binary floating point. This is not a theoretical objection:

```
0.1 + 0.2 === 0.3 → false // actually 0.30000000000000004
19.99 * 100 === 1999 → false // actually 1998.9999999999998
```

In a personal calculator that is a curiosity. In a general ledger, it compounds across millions of transactions until the trial balance will not foot, and the reconciliation is off by an amount nobody can explain because no individual transaction

looks wrong.

A second, more insidious variant: **rounding leakage in allocation**. Split one real US\$87,431.66 fuel invoice across 400 rotations. If each share is rounded independently, up to 399 cents can evaporate. The parts no longer sum to the invoice, and the difference is unallocated — so the ledger does not reconcile to the supplier statement for a reason nobody recorded.

THE SOLUTION

Three structural rules, not conventions:

1. Every monetary column is `BIGINT` named `*_cents`. No `FLOAT`. No `NUMERIC` for money.
2. All arithmetic on money lives in exactly one module — `shared/src/money.js` — which refuses to accept a float.
3. Allocation uses the **largest-remainder method** so parts sum to the total *exactly*.

SHARED/SRC/MONEY.JS — THE ONLY PLACE MONEY MATHS HAPPENS

JAVASCRIPT

```

/** Parse a human-entered amount into integer cents WITHOUT a float. */
export function cents(input) {
  const raw = typeof input === 'number' ? input.toFixed(2) : String(input);
  const cleaned = raw.replace(/[\s,]/g, '');
  const match = new RegExp('^(?)(\d*)(?:\.\d{1,2})?$').exec(cleaned);
  if (!match) throw new MoneyError('Cannot parse', 'MONEY_UNPARSEABLE');
  const [, sign, whole = '0', frac = ''] = match;
  const frac2 = (frac + '00').slice(0, 2); // pad, never round
  const value = Number(whole || '0') * 100 + Number(frac2);
  return sign === '-' ? -value : value;
}

/** Reject floats at EVERY entry point, not just at the parser. */
function assertSafeInteger(v, what = 'amount') {
  if (typeof v !== 'number' || !Number.isSafeInteger(v)) {
    throw new MoneyError(`{what} must be a safe integer number of cents; received ${v}`,
      'MONEY_NOT_SAFE_INTEGER');
  }
}

/** Split a total across weights; parts sum to EXACTLY the total. */
export function allocate(total, weights) {
  const weightSum = weights.reduce((a, b) => a + b, 0);
  const rawShares = weights.map(w => (total * w) / weightSum);
  const floors = rawShares.map(s => Math.floor(s));
  let remainder = total - floors.reduce((a, b) => a + b, 0);

  // Hand leftover cents to the largest fractional parts, biggest first.
  const order = rawShares.map((share, index) => ({ index, frac: share - Math.floor(share) })))
    .sort((a, b) => b.frac - a.frac || a.index - b.index);

  const result = [...floors];
  for (let i = 0; remainder > 0 && i < order.length; i += 1, remainder -= 1) {
    result[order[i].index] += 1;
  }
  return result; // sum(parts) === total, always
}

```

VERIFIED, NOT ASSERTED

These are shipping tests, not illustrations:

TESTS/MONEY.TEST.JS — THE FAILURE CASES THAT MOTIVATED THE DESIGN

VITEST

```
it('THE classic failure case: 0.1 + 0.2 must equal 0.30 exactly', () => {
  const a = cents('0.10'), b = cents('0.20');
  expect(subtract(sum(a, b), cents('0.30'))).toBe(0); // passes
});

it('splits a real fuel invoice across 400 flights exactly', () => {
  const invoice = cents('87431.66'); // US$87,431.66
  const weights = Array.from({ length: 400 }, (_, i) => 80 + (i % 7));
  expect(sum(...allocate(invoice, weights))).toBe(invoice); // zero leakage
});

it('rejects fractional cents from upstream callers', () => {
  expect(() => buildFlightCost({ ...f, flightCrewCents: 1850.5 }))
    .toThrow(/integer cents/);
});
```

5

P5 — Multi-tenancy is one bug away from disaster

THE PROBLEM

AMS is a SaaS product serving many airlines. In a shared database, a single missing `WHERE tenant_id = ?` exposes one carrier's fleet, fuel contracts and cost data to another. For an airline, that is not a bug report — it is the end of the business.

The dangerous part is that isolation enforced only in application code is **one refactor away from broken**. It works today, it works in every test, and it fails the first time an engineer adds an endpoint at 5pm on a Friday.

THE SOLUTION

Three independent layers. The third is the one that cannot be forgotten, because it is the database rather than the code:

1. **Middleware** — tenant id is derived from the verified JWT. It is *never* accepted from a request body or query string, so a caller cannot request another airline's data by parameter manipulation.
2. **Repository** — every statement filters explicitly.
3. **PostgreSQL RLS** — a policy on every table, so even a careless raw query with no filter returns nothing.

BACKEND/MIGRATIONS/001 — ISOLATION ENFORCED BY THE DATABASE

SQL

```
CREATE TABLE budget_lines ( ... tenant_id TEXT NOT NULL ... );

ALTER TABLE budget_lines ENABLE ROW LEVEL SECURITY;
ALTER TABLE budget_lines FORCE ROW LEVEL SECURITY;

CREATE POLICY budget_lines_tenant_isolation ON budget_lines
  USING (tenant_id = current_setting('app.tenant_id', true))
  WITH CHECK (tenant_id = current_setting('app.tenant_id', true));

-- The connection sets app.tenant_id once per transaction:
SELECT set_config('app.tenant_id', $1, true);

-- A developer who forgets the WHERE clause gets ZERO rows, not
-- another airline's fuel contracts. The database refuses.
```

ONE MORE DETAIL THAT LEAKS DATA

Cross-tenant access must return **404, not 403**. A 403 that distinguishes "exists but forbidden" from "does not exist" is itself an information leak — it lets an attacker map another carrier's fleet by registration even without reading a field. AMS asserts 404 in its adversarial suite.

6**P6 — Uplift is not burn, and nobody measures the difference****THE PROBLEM**

Costing a flight by the fuel it *uplifted* overstates that flight and understates the next one. Aircraft land with reserves — taxi fuel, contingency, alternate, final reserve, and ETOPS reserve on long-haul. That fuel is carried, not consumed, but a naive system charges it to the flight that lifted it.

Worse: **planned burn is not actual burn**. The crew files a plan from route, weight and weather; the aircraft does what it does. The variance is the single most valuable operational-financial metric an airline produces, because it isolates crew technique, routing and weight discipline from fuel *price*, which is pure market risk the airline cannot influence at all.

THE SOLUTION

Reconcile uplift to burn explicitly, carry the landing reserve forward, and separate the variance into its own cost line so management can see what is actually controllable.

WORKED EXAMPLE — SHARED/SRC/FUEL.JS, A FLIGHT SHORT OF RESERVE

JAVASCRIPT

```
const plan = { taxiFuelKg: 150, tripFuelKg: 3100, contingencyKg: 180,
  alternateKg: 220, finalReserveKg: 100,
  upliftKg: 3900, priceCentsPerKg: 95 };

const r = reconcileFuel(plan, { takeoffFuelKg: 3870, landingFuelKg: 500 });

r.reservesKg // 650 ← taxi+cont+alt+final
r.expectedLandingFuelKg // 470 ← uplift less burnable
r.reservesMeetRegulatoryMinimum // false ← SHORT BY 180 kg

r.actualLandingFuelKg // 500 ← looks fine in isolation...
r.landingReserveVarianceKg // +30 ← ...but 180 kg below requirement
r.varianceKg // +30 over plan
r.varianceCostCents // 2850 ← US$28.50 the airline could act on
```

This is a real regulatory finding, not a rounding note. A crew that consistently lands inside the final reserve has an unreported fuel-management problem, and in a fuel-expensive market it is also costing real money that is buried inside a monthly fuel line.

Tankering — a calculation, not a habit

Carrying extra fuel deliberately (tankering) trades deadweight against the risk of a diversion. Most airlines do this on instinct. AMS calculates it — and the naive formulation makes tankering look free, because it ignores the two things that matter:

TWO CORRECTIONS THE NAIVE MODEL GETS WRONG

1. **Diversion only happens sometimes.** The value of carrying contingency fuel is the *expected* cost avoided, so it must be scaled by probability of diversion. Valuing it at the gross cost of a diversion makes carrying weight appear almost always correct.
2. **Carrying fuel is not free.** It burns on every remaining sector of the rotation and may displace fuel you could uplift cheaply at the next station. That opportunity cost is real and grows with the length of the remaining rotation.

SHARED/SRC/FUEL.JS — CORRECTED TANKERING ECONOMICS

JAVASCRIPT

```
const deadweightBurnFactor = ctx.deadweightBurnFactor ?? 0.030; // kg fuel / kg payload
const probabilityOfDiversion = ctx.probabilityOfDiversion ?? 0.02;
const remainingSectors = ctx.remainingSectors ?? 3;

// Value of tankering = expected cost avoided, NOT gross diversion cost.
const expectedSavingCents = Math.round(
  probabilityOfDiversion * (ctx.costOfDiversionCents
    + ctx.fuelPriceAtAlternateCentsPerKg * plan.alternateKg));

// Cost of tankering = the fuel, plus what hauling it burns downstream.
const carryingCostCents = Math.round(
  plan.contingencyKg * ctx.fuelPriceHereCentsPerKg
    * (1 + deadweightBurnFactor * remainingSectors));

shouldTanker = expectedSavingCents > carryingCostCents;
```

7

P7 — Carbon is now a balance-sheet item, and double-counting costs millions

THE PROBLEM

Carbon is not a sustainability report any more. It is a compliance obligation with a hard number on it.

CORSIA (ICAO) requires operators above 10,000 tonnes of CO₂ a year to monitor, report and verify every international flight, then cancel emissions units against a three-year compliance period: 2021-23, 2024-26, 2027-29, 2030-32, 2033-35. New entrants are exempt for three years. Aircraft below 5,700 kg MTOM are exempt.

EU ETS has covered aviation since 2012. Allowance prices exceeded EUR 80/tonne across 2022-23, putting the sector-wide bill at roughly EUR 1.8bn in 2022 and EUR 2.1bn in 2023, and estimated at **4-6% of annual operating cost** for 2024-26. Scope has been extended to flights departing the EEA to destinations within 5,000 km.

The trap: **the two regimes overlap, and emissions must be counted exactly once.** Where both apply, CORSIA-covered tonnes are deducted from the EU ETS chargeable quantity to avoid double pricing. No mainstream library does this for you, and getting it wrong in either direction is a seven-figure error at EUR 85/tonne.

THE SOLUTION

Allocate each flight's emissions across regimes explicitly, with the deduction expressed in code and a reconciliation assertion on every flight.

SHARED/SRC/CARBON.JS — THE OVERLAP CASE, HANDLED EXPLICITLY

JAVASCRIPT

```
// A flight where BOTH regimes touch the same emissions.
allocateEmissions({
  fuelKg: 3200, international: true, corsiaCovered: true,
  eeaInvolved: true, mtomKg: 79_000,
}, { euaPriceCentsPerTonne: 8500 }); // EUR 85/tonne

{
  totalCo2Tonnes: 10.112, // 3,200 kg × 3.16 kg CO2/kg
  allocation: { CORSIA: 10.112, EU_ETS: 0, UNREGULATED: 0 },
  overlapTonnes: { deductedFromETS: 10.112 }, // ← the deduction
  allowanceCostCents: 0, // CORSIA satisfies it; not billed twice

  allocationReconciles: true, // parts reconstruct the total exactly
}
```

And this is enforced as a property across every combination of regime applicability, not just the case someone thought of:

TESTS/DOMAIN-LOGIC.TEST.JS — EMISSIONS COUNTED EXACTLY ONCE

VITEST

```
it('emissions are counted exactly once - allocation always reconciles', () => {
  const cases = [
    { international: false, corsiaCovered: false, eeaInvolved: false, mtomKg: 79_000 },
    { international: true, corsiaCovered: true, eeaInvolved: false, mtomKg: 79_000 },
    { international: true, corsiaCovered: true, eeaInvolved: true, mtomKg: 79_000 },
    { international: false, corsiaCovered: false, eeaInvolved: true, mtomKg: 79_000 },
    { international: true, corsiaCovered: false, eeaInvolved: true, mtomKg: 79_000 },
  ];
  for (const c of cases) {
    const r = allocateEmissions({ fuelKg: 5000, ...c }, ctx);
    const sumAllocated = r.allocation.CORSIA + r.allocation.EU_ETS + r.allocation.UNREGULATED;
    expect(sumAllocated).toBeCloseTo(r.totalCo2Tonnes, 6);
  }
});
```

8

P8 — An audit log the company can edit is not evidence

THE PROBLEM

An airline's audit trail is presented to a statutory auditor, to a lessor's technical representative during a return check, and to a foreign civil aviation authority inspecting the operator. If the airline can quietly edit its own history, that evidence is worth nothing.

A conventional append-only audit log is still editable by anyone holding database write access. Application-level append-only is a *policy*.

THE SOLUTION

Make tampering **detectable** rather than merely discouraged, using a cryptographic hash chain, and enforce append-only in the database itself.

BACKEND/SRC/AUDIT/CHAIN.JS — TAMPER-EVIDENT CHAINJAVASCRIPT

```
// Each entry hashes its own content TOGETHER WITH the previous hash.
export function computeHash(entry, prevHash) {
  const canonical = canonicalise({ // deterministic key order
    id: entry.id, tenantId: entry.tenantId, actorId: entry.actorId,
    entityType: entry.entityType, entityId: String(entry.entityId),
    action: entry.action, before: entry.before, after: entry.after,
    ip: entry.ip, occurredAt: entry.occurredAt,
  });
  return createHash('sha256')
    .update(prevHash, 'utf8').update(canonical, 'utf8').digest('hex');
}

export function verifyChain(entries) {
  let prev = GENESIS_HASH;
  for (const entry of entries) {
    if (computeHash(entry, prev) !== entry.entryHash) {
      return { valid: false, brokenAt: entry.id,
        reason: 'Hash mismatch: entry content or ordering has been altered' };
    }
    prev = entry.entryHash;
  }
  return { valid: true, entriesChecked: entries.length };
}
```

THE DATABASE REFUSES MUTATION OUTRIGHT — MIGRATION 001SQL

```
CREATE OR REPLACE FUNCTION ams_forbid_audit_mutation() RETURNS TRIGGER AS $$
BEGIN
  RAISE EXCEPTION 'audit_logs is append-only. entry % attempted %', OLD.id, TG_OP
  USING ERRCODE = 'restrict_violation';
END; $$ LANGUAGE plpgsql;

CREATE TRIGGER audit_logs_no_update BEFORE UPDATE ON audit_logs
FOR EACH ROW EXECUTE FUNCTION ams_forbid_audit_mutation();
CREATE TRIGGER audit_logs_no_delete BEFORE DELETE ON audit_logs
FOR EACH ROW EXECUTE FUNCTION ams_forbid_audit_mutation();
```

Verified behaviour, from the shipping suite:

Attack	Detected?	Why
Edit an amount in a 2026 entry	YES	Hash mismatch at that entry and every one after it
Delete an entry	YES	Chain breaks at the successor
Reorder entries	YES	Even with hashes intact — order is hashed
Swap the recorded actor	YES	The classic repudiation attempt
Corrupt airline A's chain	YES	Isolated to A — B still verifies

9

P9 — Airline procurement has different risk than ordinary procurement

THE PROBLEM

Generic procurement software is written for buying office furniture and IT kit. An airline buys things that can ground it:

- **Spare parts** with serialised provenance and airworthiness status. A counterfeit part in the supply chain is a catastrophic, existential risk — not a commercial one.
- **Fuel** on into-plane contracts where the volume is enormous and the price volatile.
- **Aircraft leases** under IFRS 16, which create right-of-use assets and liabilities of hundreds of millions with a defined term.
- **Ground handling and catering** priced per passenger per sector, where a route's whole margin can turn on a per-passenger cost.

THE SOLUTION

Vendor categories are aviation-specific and first-class, and every contract type carries its own obligation model rather than a generic "amount and date":

Vendor category	Special obligation AMS models
FUEL	Into-plane fee structure, uplift reconciliation per delivery note, price per kg by station and date, hedging linkage
SPARES	Serialisation, life-limited status, AD/SB applicability, shelf life, repair-versus-replace economics, counterfeit risk flags
AIRCRAFT_LEASE	IFRS 16 right-of-use asset and liability, term, residual value, return-condition obligations
GROUND_HANDLING	Per-sector and per-passenger rates, turnaround commitments, SLA breach credits
CATERING	Cost per passenger, uplift per rotation, wastage, special-meal handling
INSURANCE	Hull value tied to the asset register, deductible structure, coverage period against lease term

10

P10 — Nobody can reconcile a supplier invoice to a flight

THE PROBLEM

A fuel invoice covers every aircraft that tanked at a station in a month. A handling invoice covers every turnaround. Payroll covers every duty. At month end finance must allocate all of it across flights, and the allocation is a *policy choice* that nobody documented — so when the fleet manager disagrees with the fuel number by 4%, there is no answer to give.

THE SOLUTION

Attach allocation at the point of creation, and make the driver explicit rather than implicit. When the disagreement arises, the rule is in the system and can be pointed at.

- **BLOCK_HOURS** — cost that scales with flying time
- **ASKS** — scales with capacity offered
- **SECTORS** — scales with movements
- **DISTANCE_KM** — scales with distance flown
- **FLIGHTS** — flat per rotation
- **DIRECT** — already flight-specific, no allocation performed

Choosing wrongly does not corrupt anything — it makes the P&L *explainable* instead of *arbitrary*, which is the actual requirement.

PART III

Glossary of Meanings

3.1 How to read this glossary

This glossary is written for people who do not work in aviation. Each entry gives the term in ordinary language first, then how AMS treats it. Nothing here requires aviation knowledge to read.

THE FIVE WORDS THAT EXPLAIN MOST OF AIRLINE FINANCE

Load factor — how full the aircraft was. **CASK** — what one empty seat cost per kilometre. **RASK** — what one empty seat earned per kilometre. **Fleet-fixed cost** — cost you pay whether or not you fly (the aircraft's lease). **Marginal cost** — what one extra passenger actually adds.

A route earns money when RASK exceeds CASK. Everything in this glossary exists to make those five numbers trustworthy.

3.2 Glossary — A to C

Accountable Manager

A named individual an airline is legally required to appoint for a specific function — flight operations, technical, safety, quality or security — who personally answers for it. Civil aviation regulations in most jurisdictions specify exactly which posts must exist and who may hold them. *In AMS*: a user flagged `is_accountable_manager` with an array of approved roles, referenced when a regulator inspects the safety management system. *Civil Aviation Regulations; ICAO Annex 19.*

Air Operator Certificate (AOC)

The licence a government authority grants to an airline to operate commercially. It names which aircraft types the airline may fly and under what conditions, and it is the airline's licence to exist as an operator. Losing it ends the business. *In AMS*: stored on the tenant with its number, validity dates and issuing certification body; expiry is a visible compliance item. *ICAO Annex 6, Appendix 6.*

Airworthiness Directive (AD)

A mandatory instruction issued by a civil aviation authority to correct a safety issue across a fleet — often following an incident. It is law, not advice, and non-compliance grounds the aircraft. *In AMS*: tracked on a compliance track entirely separate from Service Bulletins, with applicability by aircraft type and serial range and a due threshold in cycles, hours or calendar. *ICAO Annex 6.*

ASKS — Available Seat Kilometres

Total passenger capacity offered, across the whole operation. Seats sold × distance flown. It is the supply side of the business and is completely unaffected by whether anyone bought a ticket. *In AMS*: `metrics.js → ask(seats, distanceKm)`, an integer, used as the denominator for every unit-cost metric. *IATA traffic metrics.*

Assignment of Benefit

An instrument redirecting the benefits of a contract to a named party. It lets an airline place an order with a vendor that invoices a different legal entity — common where a parent company buys and the subsidiary operates. *In AMS*: contract attribute on purchase orders, defaulting to the buying legal entity. *Civil aviation operational finance practice.*

ATA — Air Transport Association

The US trade association for airlines, whose reporting definitions (ATA Spec 8) are the reference standard for airline financial reporting. Where an airline has no statutory reporting regime, ATA definitions are the fallback. *In AMS*: the cost taxonomy is IATA-aligned, which is materially compatible with ATA's expense categories. *ATA Spec 8; IATA cost taxonomy.*

Base

The home airport of a crew member, which determines which duties they can be rostered onto and how much they are paid. *In AMS: Phase 2 crew scheduling; cost-relevant now because crew cost is allocated per flight. ICAO Annex 1; airline crew agreements.*

Bid Bond / Bid Security

A guarantee from a bank or approved institution that a bidder will honour its offer. Prevents bidders submitting prices they have no intention of paying. *In AMS: `bid_securities` with instrument type, issuing institution, value, validity, release and administration fee. Public procurement regulations.*

Block Hour

Time from closing the aircraft doors before departure to opening them on arrival. It is *not* airborne time — it includes taxiing and turnaround at the other end. It is the industry's standard measure of aircraft utilisation. *In AMS: the default allocation driver for fleet-fixed cost, because lease and insurance cost scale with time the aircraft exists in service. IATA; industry standard.*

Break-even Load Factor

The load factor a flight must achieve to cover its own cost. Calculated as $CASK \div RASK$. If the break-even is 82% and you are averaging 74%, you are losing money on every rotation. *In AMS: `breakEvenLoadFactorPpm()`, returned in parts-per-million so it is an exact integer. Derived from IATA metrics.*

CASM / CASK

Cost per available seat mile / kilometre — the unit cost of flying one seat, occupied or not. The single most-used efficiency metric in aviation. Fuel may be excluded to give a comparable "ex-fuel" figure. *In AMS: `caskMicrocents()`, returned in microcents-per-ASK because a cents figure rounds to zero on long-haul. IATA; BTS Form 41.*

CAVOK

"Ceiling and Visibility OK" — an operational condition under which a flight may be dispatched. Recorded because fuel planning margins differ by condition. *In AMS: fuel plan attribute affecting reserve requirements. ICAO Annex 6.*

Check Interval

How often a scheduled maintenance inspection must occur — expressed in flight hours, flight cycles, calendar months, or a combination. Which counter runs out first is a property of the specific aircraft's utilisation, not of the interval definition. *In AMS: `checkDue()` returns hours remaining, cycles remaining, the controlling counter, and days remaining at current utilisation. ICAO Annex 6; maintenance programme.*

Commercial Air Transport (CAT)

Operating an aircraft for the carriage of passengers or cargo for payment. It is the specific activity for which an AOC is required, and it triggers the full regulatory apparatus: operations manual, safety management system, airworthiness system. *In AMS: the licence scope recorded on the tenant. ICAO Annex 6.*

Cost Centre

The organisational unit that spends money — a division, department or cost centre. It forms a tree from the company down to the individual team, and every transaction must belong to exactly one. *In AMS: `cost_centers` with `parent_id`; the organisational dimension of every budget line. IFRS / management accounting.*

CORSIA

Carbon Offsetting and Reduction Scheme for International Aviation — the ICAO global scheme stabilising international aviation CO₂. Operators above 10,000 tonnes a year must monitor, report and verify every international flight and cancel emissions units on a three-year cycle. *In AMS: `carbon.js` — obligation determination, compliance-period mapping and emissions allocation. ICAO Assembly Resolution A41-22; Annex 16 Vol IV.*

Cost Per Flight

The total cost of operating one specific sector, combining everything that flight consumed. It is the atom of airline economics: every fleet decision, every route decision, is ultimately an argument about cost per flight. *In AMS: the primary output of the costing engine, computed on the day of operation. Activity-based costing applied to aviation.*

Cost Taxonomy

The controlled list of categories cost can be classified into. Without a shared taxonomy two carriers' CASK figures are not comparable, because one may include ground handling and the other may not. *In AMS: `COST_CATEGORIES` in `domain.js`, IATA-aligned, with each category structurally classified as flight-attributable, fleet-fixed or period. IATA cost taxonomy.*

CRM — Cabin Resource Management

Crew training and procedures for managing a cabin emergency — evacuation, fire, decompression. Mandatory, recurrent, and recorded per crew member. In AMS: Phase 2 training module; certification expiry is already modelled on the user record. ICAO Annex 1; Annex 6.

Cycle

One aircraft flight from brakes-off to brakes-on, irrespective of how long it took. A short hop and a long-haul each count as one cycle, which is why cycle-based maintenance is cheap for long-haul airlines and expensive for short-haul ones. In AMS:

`aircraft.current_cycles`; the counter behind LLP and AD limits. ICAO Annex 6.

Cybersecurity / CORSIA confusion warning

Note: CORSIA has nothing to do with cybersecurity. It is Carbon Offsetting and Reduction Scheme for International Aviation. See CORSIA above.

3.3 Glossary — D to H

Deadhead

A positioning flight — carrying crew and aircraft but no revenue passengers. It burns fuel, takes a slot and wears tyres, and earns nothing directly. In AMS: a flight with `passengers = 0` but full direct cost; it appears in the fleet-fixed allocation base because the aircraft was occupied. Industry standard.

Dispatch (or Flight Release)

The formal authority to operate a flight, given only after verifying aircraft airworthiness, crew legality and duty, weather, fuel and alternate sufficiency. It is a legal act, not an administrative one. In AMS: Phase 2 OCC; the airworthiness gate it depends on is already implemented and blocks dispatch. ICAO Annex 6.

Distribution

How the airline sells — direct web, call centre, travel agents, global distribution systems. It carries a commission cost per booking, and its structure determines the airline's yield and load factor. In AMS: cost category `distribution`, fleet-fixed, allocated by ASK; GDS and BSP integration is Phase 2. IATA distribution.

Diversion

Changing destination because the intended airport is unavailable — weather, closure, or fuel. Cost is not only the extra flying; it includes hotel, meals, crew duty extension and, critically, delay propagation. In AMS: probability and cost modelled explicitly in tankering economics. ICAO Annex 6; IATA disruption data.

ETOPS / EDTO

Extended-range Twin-engine Operations: the ability to fly long over-water routes with only two engines, by carrying enough reserve fuel for a diversion at any point. ETOPS-180 and ETOPS-330 denote the diversion distance certified. In AMS: the ETOPS reserve is a distinct line in the fuel plan, not folded into contingency. ICAO Annex 6, Attachment C.

EDTO Fuel

Extra fuel required to carry for long-haul twin-engine operations, calculated to a diversion destination at cruise speed. Frequently thousands of kilograms, and frequently the difference between a feasible and infeasible direct routing. In AMS: `etopsReserveKg` in the fuel plan. ICAO Annex 6.

Final Reserve

The regulatory minimum fuel remaining at landing — typically 30 minutes' flight at holding speed. It is not negotiable and it is why fuel planning is a legal obligation. In AMS: a separate component in `reconcileFuel()`, checked against landing fuel on every flight. ICAO Annex 6.

Fleet-Fixed Cost

Cost that is incurred whether or not a particular aircraft flies — lease, insurance, spares holding, salaried crew, distribution. It must be allocated to flights to make unit cost meaningful, and the choice of driver is a declared policy. In AMS:

`allocateFleetFixed()` with six explicit drivers and zero-cent leakage. Management accounting; industry practice.

Flight Folio

The legal record of a flight — times, fuel, crew, delays, technical details — used for noise and emissions complaints, accident investigation and regulatory reporting. It is a legal document with a defined retention. In AMS: flight events table, feeding both emissions and technical records. ICAO Annex 16 Vol I.

Fulfilment / Flight Leg

One sector of a journey — an aircraft from gate to gate. A rotation is the complete duty of one aircraft and crew pair, usually several legs. In AMS: the unit of cost attribution. Everything is costed per leg, then rolled up to rotation, route, fleet and month. Industry standard.

GFSM

*Government Finance Statistics Manual, the IMF's standard for classifying and comparing public finances. It is the basis of the government PFMS specification AMS was derived from. **AMS deliberately does not use it** — airlines are commercial entities reporting under IFRS with an industry cost taxonomy, and applying a governmental statistical standard would produce CASK figures no other carrier publishes. IMF GFSM 2014; see Part V.*

GDPR / PII

Data protection law and the personal information it governs. An airline processes unusually sensitive data — travel patterns, health information, biometric border data, and payment details — because its business requires it. In AMS: national ID and licence references are encrypted columns, masked by default, with access logged and retention enforced. GDPR; national aviation data-protection rules.

3.4 Glossary — I to P

Into-Plane Fee

The premium charged above the wholesale fuel price for delivering fuel to an aircraft at an airport. It covers hydrant access, equipment, staffing and margin, and is negotiated per station. A carrier's choice of airport can be driven as much by into-plane cost as by distance or demand. In AMS: fuel price per kg is captured per delivery note, per station, per date, so into-plane cost is separable from commodity price. Industry practice; competition authorities.

IFRS 16 — Leases

The accounting standard requiring almost every aircraft lease to be recognised on the balance sheet as a right-of-use asset and a matching liability, with interest unwinding over the term. It transformed airline balance sheets and created a large new finance requirement. In AMS: `lease_type` on the aircraft register, with right-of-use asset linkage and lease schedule. IFRS 16; IASB.

IOSA

IATA Operational Safety Audit — the industry's most respected independent safety and quality assessment, effectively a passport for IATA membership. In AMS: quality management module will hold IOSA evidence; the safety record structure it requires is modelled. IATA IOSA programme.

Load Factor

The percentage of available seats actually sold. It is the single strongest driver of profitability, because most of an aircraft's cost is fixed and the marginal cost of an extra passenger is small. In AMS: `loadFactorPpm()`, computed in parts-per-million, with a hard rejection if passengers exceed seats. IATA traffic metrics.

LLP — Life-Limited Part

A component with a hard, non-extendable life limit in cycles, hours or calendar months. At limit it must be destroyed, not repaired. Operate past limit and you risk your AOC. In AMS:

`llpStatus()` normalises all three counters and reports the controlling limit explicitly. ICAO Annex 6; national airworthiness rules.

Maintenance Reserve

A provision built up in advance for future major maintenance, calculated per flight hour or cycle. Matching the actual bill to the provision is a recurring reconciliation problem. In AMS:

cost category `maintenance_reserve`, flight-attributable, provisioned per block hour. IFRS; airline accounting practice.

MEL — Minimum Equipment List

The list of items that may be inoperative on a flight, each with an approved repair interval — a fixed number of days, a fixed number of calendar months, or flight hours. Deferring past the limit is itself a violation. In AMS:

`melStatus()` tracks CAT B repair intervals, indefinite deferrals and expiry. ICAO Annex 6, Attachment E.

MTOM

Maximum Take-Off Mass — the heaviest the aircraft may legally be. Used for weight and balance, payload and fuel planning, and as the threshold below which both CORSIA and EU ETS exempt an aircraft entirely. In AMS:

`aircraft.mtom_kg`; the 5,700 kg exemption threshold is checked on every emissions calculation. ICAO; EU ETS Directive.

Marginal Cost

The cost of one additional unit of output, holding everything else constant. For an airline, the marginal cost of one extra passenger is fuel to lift their weight plus their meal — not another landing fee or another cabin crew member. In AMS:

`estimateMarginalSeatCost()`. This number determines whether filling an empty aircraft is profitable. Microeconomic theory applied to aviation.

NPS — Net Promoter Score

A customer loyalty metric. Relevant because cargo and belly revenue depends on holding a booking relationship, not merely completing one flight. In AMS:

Phase 2 commercial module. Industry standard.

O&D — Origin and Destination

A passenger who begins and ends their journey at the airports in question. It is the traffic an airline genuinely controls, as opposed to a connecting passenger the airline merely carried. O&D pricing and O&D load factor are the meaningful measures of route quality. In AMS:

Phase 2 revenue attribution; the costing engine already separates flight-leg economics from passenger-level revenue. IATA traffic metrics.

Operations Manual

The controlled document set describing how the airline operates — aircraft, crews, weather, MEL, emergency and maintenance procedures. Its approval is a condition of the AOC. In AMS:

every procedure referenced by the platform maps to an operations manual section identifier. ICAO Annex 6, Attachment G.

Period Cost

Cost that is not caused by flying — head office, IT, financing, income tax, non-aircraft depreciation. Excluded from unit cost because no board decides a route based on it. In AMS:

`PERIOD_CATEGORIES`, structurally excluded from CASM. Getting this boundary wrong is the commonest reason two carriers' CASK figures are incomparable. IFRS; IATA cost taxonomy.

Pipeline Reservation

Budget capacity held for a requisition that has been submitted but not yet approved. Without it, an approver sees headroom that fifteen pending requests already consumed. In AMS:

`pending_cents` subtracted from availability at submission, released on rejection or cancellation. Funds management principle; see Part II P2.

Procurement Plan

A published statement of what the airline intends to buy in a period, with values, funding sources and lead times. It makes procurement visible before it happens and is a statutory obligation in many jurisdictions. In AMS:

`procurement_plans` with plan lines; a sourcing event cannot be published without a linked approved plan line. Public procurement regulations.

PVPC — Production Variance Passenger Capacity

The gap between expected capacity and actual capacity produced, caused by cancellations, gauge changes or load-shedding. It is a direct revenue and cost variance. In AMS:

reported by sector in the executive dashboard. IATA; airline planning.

3.5 Glossary — Q to Z

RAF / RASK

Revenue per available seat kilometre — total revenue divided by capacity offered. Compared against CASK it determines whether a route earns money. In AMS:

`raskMicrocents()`. IATA traffic metrics.

Revenue Accounting

How an airline recognises what it earned from a flight — passenger fares, ancillaries, cargo belly revenue, and interline settlements. Complex because ticket revenue is collected in one currency, earned in another, and settled through clearing houses months later. In AMS:

revenue attributed per flight; interline and BSP settlement is Phase 2. IFRS 15; IATA settlement.

RNAV / RNP

Area navigation and Required Navigation Performance — precision navigation capability, which permits more direct routings and reduces fuel burn and track miles. Has direct cost consequences and increasingly grants a small amount of overfly fee exemption. In AMS:

navigation cost is captured per sector; route optimisation is Phase 2. ICAO PANS; IRNAV.

Rotation

The complete assignment of one aircraft and one crew pair for several legs over multiple days, ending back at base. It is how aircraft and crew are actually scheduled, and the real unit of crew and aircraft cost. In AMS:

flights roll up to rotations, then to routes. Industry standard.

RVSM

Reduced Vertical Separation Minimums — the ability to fly safely with reduced vertical spacing in specific airspaces, saving fuel and increasing track capacity. Requires crew qualification and aircraft approval. In AMS:

operational certification attribute. ICAO Annex 6, Appendix 4.

Safety Management System (SMS)

The formal, risk-based framework through which an airline manages safety — identifying hazards, assessing risk, mitigating, and monitoring. It is a condition of holding an AOC. In AMS:

Phase 2 safety module; the occurrence and risk register structure is defined. ICAO Annex 19; SMS Framework.

SB — Service Bulletin

The manufacturer's instruction describing a modification, inspection or improvement. Often the technical substance of a later AD — but **until an authority adopts it as an AD, applying the SB alone does not make the operator compliant**. In AMS:

tracked on a separate compliance track from ADs, with `requiresMandatoryInstruction` exposed. ICAO Annex 6; see Part II P3.

Segmentation (IFRS)

Breaking the airline's revenue into meaningful categories — by route, geography, customer type and service — so a reader can judge performance and predict cash flows. In AMS:

cost and revenue dimensions include route, fleet type and division, producing route-level P&L directly. IFRS 8.

Settlement (BSP)

The Billing and Settlement Plan — the IATA clearing system through which agents' payments and airlines' collections are netted, so the airline does not chase small balances per ticket. In AMS:

Phase 2; the receivables model is designed to accommodate it. IATA BSP.

Slot

The right to depart or arrive at a congested airport during a specific period. Slots are scarce, tradable and allocated by a coordinator — often the airport itself or a state body — and are frequently more valuable than the route. In AMS:

ground operations module Phase 2; slot cost is captured as airport charges. IATA Slot; IATA SSIM/SCR messaging.

SSIM

Standard Schedules Information Message — the standard format for exchanging airline schedules. A carrier's whole published timetable travels as one of these files. In AMS:

Phase 2 schedule module; the ingestion boundary is defined. IATA SSIM format.

Stage Length

The distance of a route. Cost per seat rises with stage length because fuel and crew duty scale with time — but revenue usually does not. This asymmetry is why long-haul has higher CASK but is not necessarily unprofitable. In AMS:

`distanceKm` per flight, the denominator of ASK. IATA traffic metrics.

Turnaround

Everything between one aircraft arriving and departing again: cleaning, refuelling, catering, boarding, baggage, maintenance checks. It is measured ground time and it directly determines how many legs an aircraft flies per day. In AMS:

turnaround costs land in handling, de-icing and crew ground duty. IATA; operations research.

Utilisation

Block hours flown per aircraft per day. Every fixed cost in the business is divided by this number, so it is the single most powerful lever on unit cost. In AMS:

`utilisationMilli()`, and the basis for maintenance check planning. IATA; airline operations.

Working Capital

Money tied up in receivables, inventory and prepayments. For an airline it can be substantial: crews are paid in advance of flying, and fuel suppliers extend terms while agents pay later. In AMS:

`finance_cost` category; period cost, excluded from CASK. IFRS; corporate finance.

Yard Time

Time an aircraft spends at a maintenance facility. Expensive and directly revenue-reducing — a planned visit is cheap and schedulable; an unplanned one is neither. In AMS:

maintenance module; predictive maintenance (Part VII F8) exists to shift cost from the second kind to the first. Industry practice.

Yield

Average revenue per passenger, as opposed to per available seat. Yield falls when an airline discounts to fill an aircraft. High load factor with low yield can be worse than moderate load factor with high yield. In AMS:

Phase 2 revenue management; per-flight revenue is already captured so yield is derivable. IATA traffic metrics.

3.6 Glossary — engineering terms in this codebase

Integer minor units

Storing money as a whole number of cents (or pence) instead of a decimal. Eliminates floating-point rounding entirely. In AMS: every monetary column is `BIGINT` named `*_cents`. See Part II P4.

Row lock (`SELECT ... FOR UPDATE`)

A database instruction that takes an exclusive lock on a row until the transaction ends, blocking concurrent readers who request the same lock. It converts a race condition into a queue. In AMS: the mechanism that makes double-spend impossible. PostgreSQL; see Part II P2.

Generated column

A column whose value is computed by the database from other columns, and which cannot be written directly. It can never disagree with its inputs. In AMS: `available_cents`, so availability cannot drift from its components. PostgreSQL 12+; SQL standard

GENERATED ALWAYS AS.

Row-Level Security (RLS)

A database feature that filters rows automatically according to a policy, regardless of what the query asks for. A query that forgets its filter returns nothing rather than everything. In AMS: the third and unforgettable layer of tenant isolation. PostgreSQL RLS; see Part II P5.

Hash chain

Storing a cryptographic hash of each record together with the previous record's hash, so any historical alteration breaks every subsequent link. In AMS: the audit log; tamper-evident rather than tamper-proof. See Part II P8.

Largest-remainder allocation

An apportionment method that distributes a whole number across parts so the parts sum exactly to the original, by giving the leftover units to the largest fractional remainders. In AMS: `allocate()`, guaranteeing zero rounding leakage across hundreds of flight allocations. See Part II P4.

Microrents / microcents

Units one millionth of a currency unit. Used for cost-per-kilometre metrics, where expressing a value in whole cents would round to zero and destroy the signal. In AMS: CASK and RASK are returned in microcents per ASK — still exact integers. See Part XII.

Parts per million (ppm)

A ratio expressed as units per million — the integer form of a percentage. Used for load factor, burn variance, FX rates and percentage allocations, so ratios can be stored and compared without float drift. In AMS: 850,000 ppm = 85.00%. Common engineering practice.

checkJs

A TypeScript compiler mode that type-checks JavaScript files using JSDoc annotations and emits no output. Used purely as a CI gate. In AMS: ADR-001 — found 47 genuine defects on first run, including two source bugs the unit tests missed. See Part XIII.

Outbox pattern

Writing a message to a table in the same transaction as the business change, then delivering it asynchronously. Guarantees a side effect is never lost when the business change commits. In AMS: every email, SMS, bank file and carbon report originates in `outbox`. Enterprise integration pattern.

Zod

A schema library where one declaration serves as both runtime validator and type source. In a JavaScript codebase it is the primary defence against malformed or malicious input. In AMS: every request boundary is parsed before any business logic runs. Community standard.

PART IV

Index of Abbreviations

Every abbreviation used in this document and in airline finance, in one alphabetical index. The "what it is" column is written for a reader who has never met the term.

N/A

Abbr.	Expansion	What it is, in plain language	Where it appears
A&C	Air Conditioning	Cost line for aircraft environmental systems	Costing
AD	Airworthiness Directive	A mandatory safety instruction from the civil aviation authority	Fleet
ADC	Air Data Computer	The flight management computer	Fleet
ADL	Approved Data List	The single authoritative source of live technical data	Fleet
AFR	Aircraft Flight Record	The permanent log of an aircraft's life	Fleet
AGM	Annual General Meeting	Shareholder meeting; approves the annual report	Governance
AOC	Air Operator Certificate	The licence to operate an airline commercially	Regulatory
APU	Auxiliary Power Unit	The engine in the tail providing electricity and air on the ground	Fleet
ASA	Aviation Safety Advisory	A safety recommendation, unlike an AD which is mandatory	Safety
ASK	Available Seat Kilometre	Passenger capacity offered — seats × distance	Metrics
ASM	Available Seat Mile	The US-mile version of ASK	Metrics
ATA	Air Transport Association	US airline trade body; its reporting spec is the industry fallback standard	Reporting
ATC	Air Traffic Control	The service that sequences aircraft in the sky	Ops
ATPL	Airline Transport Pilot Licence	The highest commercial pilot licence, allowing command	Crew
ATS	Air Traffic Services	ATC, flight information, and weather services	Ops
B2B	Business to Business	Corporate travel sales channel	Commercial
B2C	Business to Consumer	Direct retail sales channel	Commercial
BI	Business Intelligence	Reporting and analytics layer	Reporting
BPM	Bill of Materials / Business Process Management	Component list; process orchestration	Procurement
BSP	Billing and Settlement Plan	IATA's system for netting agent payments	Commercial
CAAC	Civil Aviation Administration of China	A national aviation authority	Regulatory
CAAZ	Civil Aviation Authority of Zimbabwe	The national aviation authority for Zimbabwe	Regulatory

<i>Abbr.</i>	<i>Expansion</i>	<i>What it is, in plain language</i>	<i>Where it appears</i>
CAVOK	<i>Ceiling and Visibility OK</i>	<i>Conditions permitting dispatch</i>	<i>Ops</i>
CASM	<i>Cost per Available Seat Mile</i>	<i>Unit cost of flying one seat one mile — the core efficiency metric</i>	<i>Metrics</i>
CASK	<i>Cost per Available Seat Kilometre</i>	<i>Same, in kilometres — the metric AMS uses</i>	<i>Metrics</i>
CAT	<i>Commercial Air Transport</i>	<i>Carrying passengers or cargo for payment — the activity needing an AOC</i>	<i>Regulatory</i>
CDL	<i>Configuration Deviation List</i>	<i>Approved optional equipment that may be missing or inoperative</i>	<i>Fleet</i>
CFR	<i>Code of Federal Regulations</i>	<i>US aviation regulation</i>	<i>Regulatory</i>
CMA	<i>Crew Meal Allowance</i>	<i>Per-diem cost for crew meals</i>	<i>Crew</i>
CMS	<i>Crew Management System</i>	<i>Software that builds and maintains legal rosters</i>	<i>Phase 2</i>
COA	<i>Certificate of Airworthiness / Chart of Accounts</i>	<i>Airworthiness document; or the account listing</i>	<i>Fleet / Finance</i>
CoA	<i>Certificate of Airworthiness</i>	<i>Proves the aircraft is legally airworthy</i>	<i>Fleet</i>
CORSIA	<i>Carbon Offsetting and Reduction Scheme for International Aviation</i>	<i>ICAO's global scheme; operators > 10,000 tCO₂/yr must offset</i>	<i>Carbon</i>
COSTL	<i>Cost Limit of an Airplane</i>	<i>The fuel amount at which the aircraft must stop, for risk planning</i>	<i>Ops</i>
CRM	<i>Cabin Resource Management</i>	<i>Crew training for cabin emergencies</i>	<i>Safety</i>
CRM (cost)	<i>Customer Relationship Management</i>	<i>Sales and service software — note the acronym is ambiguous in aviation</i>	<i>Commercial</i>
CTK	<i>Cargo Tonne Kilometre</i>	<i>Actual freight traffic — the cargo equivalent of RPK</i>	<i>Metrics</i>
CLF	<i>Cargo Load Factor</i>	<i>CTK ÷ ACTK — how full the freight hold is</i>	<i>Metrics</i>
CU	<i>Cost Unit / Cost per Unit</i>	<i>A generic unit-cost denominator used in airline reporting</i>	<i>Metrics</i>
DB	<i>Database</i>	<i>The system of record</i>	<i>Technology</i>
DCI	<i>Departure Control Inspector</i>	<i>Airport staff handling departing flights</i>	<i>Ground</i>
DER	<i>Descent</i>	<i>Arrival operating cost element</i>	<i>Costing</i>
DGR	<i>Dangerous Goods Regulations</i>	<i>IATA rules for transporting hazardous cargo</i>	<i>Cargo</i>
DMAS	<i>Dangerous Goods Management System</i>	<i>Shipper's obligations under the DGR</i>	<i>Cargo</i>
EDTO	<i>Extended-range Twin-engine Operations</i>	<i>Flying long over-water routes with two engines</i>	<i>Ops</i>
EFB	<i>Electronic Flight Bag</i>	<i>Tablet-based flight manuals and charts</i>	<i>Cockpit</i>
EGNOS	<i>European Geostationary Navigation Overlay Service</i>	<i>Satellite augmentation to GPS</i>	<i>Nav</i>
EHM	<i>Engine Health Monitoring</i>	<i>Continuous engine data capture for predictive maintenance</i>	<i>Maintenance</i>

Abbr.	Expansion	What it is, in plain language	Where it appears
EOI	Expression of Interest	Preliminary invitation to bid, used to shortlist suppliers	Procurement
EPUB	Electronic Publishing and Distribution	Digital distribution of manuals and circulars	Regulatory
ETA	Electronic Travel Authorisation	Advance passenger screening clearance	Security
ETOPS	Extended-range Twin-engine Operations	Same as EDTO — the term varies by region	Ops
EU ETS	European Union Emissions Trading System	Carbon pricing covering flights in and out of the EEA; 4-6% of airline operating cost	Carbon
FAA	Federal Aviation Administration	US aviation authority	Regulatory
FAI	First Aid and Initial Treatment	Cabin crew medical qualification	Crew
FAR	Federal Aviation Regulations	US aviation rules	Regulatory
F&B	Food and Beverage	Catering operations and cost	Costing
FIT	Flexible Labour and Resource Planning	Crew roster optimisation technique	Phase 2
FL	Flight Level	Barometric altitude band flown by rule of thumb	Ops
FO	First Officer	The pilot in command's deputy	Crew
FPA	Flight Planning Application	Software producing the operational flight plan	Ops
FRMS	Fuel Management System (System) / Fatigue Risk Management	Electronic fuel planning; or fatigue safety management	Ops / Safety
FTE	Full-Time Equivalent	A standardised staffing measure	HR
FTS	Flight Technical Supervisor	Overseeing flight crew technical standards	Crew
FX	Foreign Exchange	Currency conversion — a major airline cost and loss source	Finance
GASP	Gate Acceptance System Protocol	Message format for slots	Ground
GDS	Global Distribution System	Amadeus, Sabre, Travelport — where travel agents book	Phase 2
GFSM	Government Finance Statistics Manual	IMF public-finance classification — used by the government PFMS, deliberately NOT by AMS	Finance
GHG	Greenhouse Gas	Emissions accounting term	Carbon
GS	Ground Service	Ramp handling and turnaround services	Ground
HAC	Hours of Aircraft Availability	Aircraft utilisation measure	Metrics
HCA	Handover and Briefing Checklist	Crew-to-crew transfer procedure	Crew
HUB	Connecting Hub	A bank at a single airport — historically the dominant network model	Network
IA	IATA Operational Audit / Internal Audit	Industry audit; or finance assurance	Audit
IATA	International Air Transport Association	The airline trade body; defines most industry formats	Industry

Abbr.	Expansion	What it is, in plain language	Where it appears
<i>IATF</i>	<i>International Aviation Training Forum</i>	<i>Sets training standards</i>	<i>Training</i>
<i>ICAO</i>	<i>International Civil Aviation Organization</i>	<i>The UN body whose Annexes are global aviation standards</i>	<i>Regulatory</i>
<i>ICC</i>	<i>Interline and Clearing Credit</i>	<i>Settlement amount owed between airlines</i>	<i>Commercial</i>
<i>IFRS</i>	<i>International Financial Reporting Standards</i>	<i>Accounting rules that govern airline financial statements</i>	<i>Finance</i>
<i>IKS</i>	<i>Inflight Kitchen Services</i>	<i>Catering supplier term in the Middle East</i>	<i>Costing</i>
<i>IMM</i>	<i>Inflight Maintenance Manual</i>	<i>How minor defects are handled without a full hangar visit</i>	<i>Maintenance</i>
<i>INM</i>	<i>Interface Control Document</i>	<i>Specification for a system-to-system exchange</i>	<i>Integration</i>
<i>IOSA</i>	<i>IATA Operational Safety Audit</i>	<i>The industry's benchmark safety and quality audit</i>	<i>Safety</i>
<i>JIT</i>	<i>Just in Time</i>	<i>Inventory arriving exactly when needed — reduces warehouse cost</i>	<i>Supply</i>
<i>KPI</i>	<i>Key Performance Indicator</i>	<i>A tracked measure of performance</i>	<i>Reporting</i>
<i>KPIh</i>	<i>Key Performance Indicator per Hour</i>	<i>A rate normalised by flight hours</i>	<i>Reporting</i>
<i>KPI PK</i>	<i>Key Performance Indicator per Passenger-Kilometre</i>	<i>A rate normalised by revenue distance</i>	<i>Reporting</i>
<i>KPI PL</i>	<i>Key Performance Indicator per Payload-Kilometre</i>	<i>A rate normalised by weight distance</i>	<i>Reporting</i>
<i>KPI FT</i>	<i>Key Performance Indicator per Flight-Turn</i>	<i>A rate normalised by flights</i>	<i>Reporting</i>
<i>LDG</i>	<i>Landing</i>	<i>Arrival operating element</i>	<i>Costing</i>
<i>LHR</i>	<i>Heathrow</i>	<i>The most slot-constrained airport in the world</i>	<i>Network</i>
<i>LLP</i>	<i>Life-Limited Part</i>	<i>A component with a hard, non-extendable life limit</i>	<i>Fleet</i>
<i>LRU</i>	<i>Line Replaceable Unit</i>	<i>A part swapped at the flight line without workshop time</i>	<i>Maintenance</i>
<i>LT</i>	<i>Long Term</i>	<i>Long-haul; or a long-term contract</i>	<i>Contract</i>
<i>LTCC</i>	<i>Life-Time Cost</i>	<i>Total cost of ownership over an asset's life</i>	<i>Finance</i>
<i>LZ</i>	<i>Loading Zone</i>	<i>Where an aircraft is loaded</i>	<i>Ground</i>
<i>MEL</i>	<i>Minimum Equipment List</i>	<i>What may be inoperative and for how long</i>	<i>Fleet</i>
<i>MH</i>	<i>Message Header</i>	<i>IATA message format</i>	<i>Integration</i>
<i>MRV</i>	<i>Monitoring, Reporting and Verification</i>	<i>The backbone of CORSIA compliance</i>	<i>Carbon</i>
<i>MSN</i>	<i>Manufacturer Serial Number</i>	<i>An aircraft's unique identity at the factory</i>	<i>Fleet</i>
<i>MTOM</i>	<i>Maximum Take-Off Mass</i>	<i>Heaviest legal weight; the 5,700 kg carbon exemption threshold</i>	<i>Fleet</i>

Abbr.	Expansion	What it is, in plain language	Where it appears
MWM	Minimum Wet Mass	Lightest weight at which the aircraft may be dispatched	Ops
Not Available / Not Applicable	Standard reporting notation	Reporting	
NPA	New Passenger Announcement	New route public announcement	Commercial
OCA	Operational Conversion Agreement	Contract trading an aircraft without operating it	Finance
OCC	Operations Control Centre	The room where the day's operation is managed and disrupted	Ops
O&D	Origin and Destination	A passenger who began and ended at your airports	Metrics
OIL	Operations Integrity Level	IATA route quality classification	Industry
OPEM	Operational Profitability Evaluation Model	Fast per-flight cost and revenue calculation	Costing
OPW	Operating Weight	Actual aircraft weight at a point	Ops
PAX	Passengers	Passenger count on a flight	Metrics
PDF	Portable Document Format	Report output format	Technology
PLF	Passenger Load Factor	$RPK \div ASK$ — the percentage full	Metrics
PNR	Passenger Name Record	The booking record for one passenger	Phase 2
PPA	Purchase Price Allocation	How a business combination is split across the balance sheet	Finance
PVPC	Production Variance Passenger Capacity	Capacity lost versus planned	Metrics
RA	Revenue Accounting	How airline revenue is earned and recognised	Commercial
RAAT	Revenue Accounting System	The software doing it	Commercial
RAM	Resource Allocation Management	Assigning scarce assets to flights	Planning
RASM	Revenue per Available Seat Mile	Unit revenue in US miles	Metrics
RASK	Revenue per Available Seat Kilometre	Unit revenue — the metric AMS uses	Metrics
RFP	Request for Proposals	Procurement for complex services	Procurement
RFQ	Request for Quotation	Procurement of quoted prices for low-value needs	Procurement
RGS	Reusable Glyph Storage	Load-balancing component used in web tiering	Technology
RNAV	Area Navigation	Navigation between waypoints rather than by ground tracks	Nav
RNP	Required Navigation Performance	Navigation accuracy assurance	Nav
ROT	Rotation	Aircraft and crew assignment over several days	Crew
RRP	Revenue Recognition Point	When revenue is booked	Finance
RSM	Revenue Sharing Model	Distribution of interline revenue	Commercial

<i>Abbr.</i>	<i>Expansion</i>	<i>What it is, in plain language</i>	<i>Where it appears</i>
<i>RSP</i>	<i>Required Service Performance</i>	<i>Minimum in-flight service standard</i>	<i>Commercial</i>
<i>RTGS</i>	<i>Real Time Gross Settlement</i>	<i>The central high-value interbank payment rail</i>	<i>Finance</i>
<i>RVSM</i>	<i>Reduced Vertical Separation Minima</i>	<i>Permitting tighter vertical spacing in certain airspace</i>	<i>Ops</i>
<i>SAC</i>	<i>Specialised Airport Clearance</i>	<i>Approval to operate into a restricted aerodrome</i>	<i>Regulatory</i>
<i>SAF</i>	<i>Sustainable Aviation Fuel</i>	<i>Non-fossil jet fuel, eligible for a zero emissions factor</i>	<i>Carbon</i>
<i>SAEZ</i>	<i>Scheduled Agreement for EU ETS / Zone</i>	<i>Regulatory wording</i>	<i>Carbon</i>
<i>SAT</i>	<i>Seat Assignment Technique</i>	<i>Cabin seating optimisation</i>	<i>Commercial</i>
<i>SB</i>	<i>Service Bulletin</i>	<i>Manufacturer instruction — not legally binding until adopted by an AD</i>	<i>Fleet</i>
<i>SCM</i>	<i>Supply Chain Management</i>	<i>Procurement through to delivery</i>	<i>Procurement</i>
<i>SCR</i>	<i>Standard Schedules Information Request</i>	<i>IATA slot request message</i>	<i>Ground</i>
<i>SFD</i>	<i>Sale of Flight Distance</i>	<i>IATA legacy distance metric</i>	<i>Metrics</i>
<i>SMA</i>	<i>Slot Message Application / Standards Manual of Accounting</i>	<i>Slot confirmation message; or IATA accounting rules</i>	<i>Ground / Finance</i>
<i>SMS</i>	<i>Safety Management System</i>	<i>The formal safety framework an AOC requires</i>	<i>Safety</i>
<i>SN</i>	<i>Serial Number</i>	<i>Aircraft MSN or part serial</i>	<i>Fleet</i>
<i>SOP</i>	<i>Standard Operating Procedure</i>	<i>How a specific task is performed</i>	<i>Operations</i>
<i>SPH</i>	<i>Standard Platform Load</i>	<i>Load factor used when platform data is unavailable</i>	<i>Reporting</i>
<i>SSIM</i>	<i>Standard Schedules Information Message</i>	<i>The schedule exchange file format</i>	<i>Planning</i>
<i>SSR</i>	<i>Solid State Recorder</i>	<i>The aircraft flight data recorder</i>	<i>Fleet</i>
<i>STBY</i>	<i>Standby</i>	<i>Crew on call to cover disruption</i>	<i>Crew</i>
<i>TAT</i>	<i>Turnaround Time</i>	<i>Ground time between arrival and departure</i>	<i>Ground</i>
<i>TAT-A</i>	<i>Turnaround Time — Aircraft</i>	<i>Ground time as it affects the aircraft</i>	<i>Ground</i>
<i>TAT-P</i>	<i>Turnaround Time — Passenger</i>	<i>Ground time as the passenger experiences it</i>	<i>Ground</i>
<i>TSA</i>	<i>Transport Security Administration / Treasury Single Account</i>	<i>US security screening; or a government single-bank-account structure</i>	<i>Security / Finance</i>
<i>UAS</i>	<i>Unmanned Aircraft System</i>	<i>A drone; now separately regulated by ICAO</i>	<i>Regulatory</i>
<i>UPRT</i>	<i>Unit Price of Route Ticketing</i>	<i>Aircraft lease rate per block hour</i>	<i>Finance</i>
<i>VAT</i>	<i>Value Added Tax</i>	<i>Consumption tax, charged on most airline purchases</i>	<i>Finance</i>
<i>W&B</i>	<i>Weight and Balance</i>	<i>The load distribution calculation</i>	<i>Ops</i>
<i>WOW</i>	<i>Weight of Water</i>	<i>Fuel weight on board, for W&B</i>	<i>Ops</i>
<i>XAC</i>	<i>Exceptional Conditions</i>	<i>Non-standard handling requirements</i>	<i>Ground</i>

THREE ACRONYMS IN THIS TABLE THAT GENUINELY OVERLAP

CRM means *Cabin Resource Management in a safety context and Customer Relationship Management in a commercial one. Always qualify it.*

TSA means *the US screening body in a security context and Treasury Single Account in a government-finance context. AMS uses the latter only by inheritance from the government specification.*

COA means *Certificate of Airworthiness or Chart of Accounts depending on whether it is prefixed. Typographic capitalisation is the only disambiguation, which is why it should be avoided in writing.*

PART V

Background and the Incumbent Landscape

5.1 What the market sells today

Airline systems divide into four commercial categories. None of them does what AMS does, and understanding why is the justification for building it.

Category	Representative products	Strength	Where it fails the finance question
Operational planning suites	Lufthansa Systems NetLine; PDC Airline Suite; AIMS	Schedule, slots, crew rostering, operations control — genuinely mature	They compute operational feasibility and hand a cost estimate to a finance system that reconstructs the real number separately. Two systems, two answers, no reconciliation.
Aviation add-ons to a generic ERP	Odoo + SmartOPS; SAP aviation extensions	Free flight-ops capability once you have the ERP	The ERP's costing is designed for a discrete-manufacturing cost model. Fuel, crew duty and fleet-fixed allocation do not map cleanly, so airlines bolt on spreadsheets.
Route and profitability analytics	Cirium, OAG, specialized BI vendors	Fast, credible route-level insight using published schedule data	Operates on published schedules, not on what actually flew. It cannot see an unscheduled fuel stop, an extra cabin crew hour, or a technical delay that consumed fuel. It is excellent and structurally incomplete.
Legacy airline core systems	Decades-old mainframe revenue accounting platforms	They do run real airlines	Enormous cost, slow change, closed data model. Extracting cost-per-flight to answer one question is a multi-month project. Many are so hard to replace that they define the airline.

5.2 The gap every one of them leaves

THE OBSERVATION THIS PLATFORM IS BUILT ON

All four categories answer "was this route profitable last quarter?" and none answers "what did this flight cost, and can we afford it?" — because the data required for the second question lives in operational systems, and the data required for the first lives in finance systems, and the two are joined by an accrual estimate that neither team fully trusts.

AMS is built on the opposite premise: **one record, three views**. The fuel delivery note, the crew duty report and the landing receipt are the facts. Cost per flight, route P&L and the general ledger are three renderings of those same facts at three grains. There is no join, no accrual estimate, and no opportunity for the two to disagree.

5.3 Why not start from the government specification

AMS descends from a government public financial management system specified in the same workspace. Three things were deliberately discarded:

Inherited	Why discarded in AMS
IMF GFSM chart of accounts	<i>GFSM is a statutory government statistical standard. Airlines report under IFRS with an industry cost taxonomy. Applying GFSM would produce CASK figures no other carrier publishes — technically compliant and commercially useless.</i>
Votes and sub-votes	<i>A governmental appropriation hierarchy. The airline equivalent is division / route / fleet type, which is how commercial decisions actually get framed.</i>
Treasury warrants and a Treasury Single Account	<i>State fiscal machinery. An airline has bank facilities, a Treasury function and a working-capital cycle — not a Consolidated Fund.</i>

What was **kept** and proved out by the same team: multi-tenant isolation, segregation of duties, hash-chained audit, integer money, and above all the commitment-control discipline. That control is domain-independent. A ministry and an airline both fail the same way when two people approve against the same balance at the same moment.

PART VI

Aviation Regulatory Framework

6.1 The ICAO Annex system

Almost all international aviation law comes from nineteen Technical Annexes to the Convention on International Civil Aviation. Four matter to AMS directly.

Annex	Subject	What it requires of the operator	AMS consequence
Annex 1	Personnel Licensing	Licences, type ratings, medical currency, training records	Licence and medical expiry tracked per user; recurrent training currency gates crew assignment (Phase 2)
Annex 6	Operation of Aircraft	AOC; Operations Manual; MEL; flight folio; dispatch; fuel reserves; EDTO	The single densest source of required behaviour. MEL, fuel reserves and airworthiness are implemented now.
Annex 16 Vol I / Vol IV	Noise / Environmental Protection	Fuel use per flight; CORSIA MRV	Flight-level fuel data is the substrate of CORSIA reporting. Built in, not retrofitted.
Annex 19	Safety Management	A formal SMS: hazard identification, risk assessment, mitigation, monitoring	Requires accountable managers to be named — drives the departmental structure (Part IX)

6.2 National certification

Each State licenses its own operators. Zimbabwe's authority is the **Civil Aviation Authority of Zimbabwe (CAAZ)**, which licenses aviation personnel, approves training organisations, and oversees operations. In this region, South Africa's SA-CATS and similar national instruments apply the same structure with local variations — notably a required set of appointed posts: Chief Executive Officer, person responsible for flight operations, person responsible for aircraft, Air Safety Officer, Quality Assurance Manager and Security Manager.

WHY THIS SHAPES THE PRODUCT, NOT JUST THE COMPLIANCE PAGE

Because regulations require named accountable managers, a department structure is a legal requirement rather than an organisational preference. Every one of those six posts needs their own system view: the Flight Operations manager needs legality and dispatch status, the Technical manager needs airworthiness and parts, the Safety officer needs occurrences and risk, the QA manager needs audit evidence and findings, the Finance director needs cost per flight and commitments, the CEO needs the whole picture.

That is why AMS's portal registry has 16 named departments rather than a role-permission soup. It is also why the accountable-manager appointment and its regulatory approval are first-class records.

6.3 Operational specifications

An AOC is issued with attached operational specifications that state exactly which aircraft types the operator may fly and under which conditions — including approved routes, RVSM and PBN authorisations, and low-visibility operations. AMS stores the AOC scope on the tenant and treats fleet authorisation as a constraint on operations.

6.4 Independent assessment

IOSA (IATA Operational Safety Audit) is the industry's most respected independent safety and quality assessment and effectively a condition of IATA membership. **AFCAP** (African Civil Aviation Commission) is the regional safety audit body.

Both demand documented, retrievable procedures and records — which is why AMS's audit chain and document retention are compliance infrastructure, not nice-to-haves.

PART VII

Frontier Architecture

READ THE READINESS COLUMN HONESTLY

Every capability below is rated for how ready it is to build and operate, not how exciting it sounds. **BUILT** means shipping in Wave 1. **FUNDAMENTAL** means the data model is designed and it is an implementation task. **REQUIRES VALIDATION** means it is well understood in industry but needs a prototype and expert review before commitment. **RESEARCH** means it is a live frontier with real open questions.

7.1 F1 — Digital twin of the cost and fleet model

BUILT · **FUNDAMENTAL**

Today's costing engine answers "what did this flight cost". A twin answers "what will it cost" — for a flight not yet flown, under specific weight, weather, routing and fuel assumptions — and, crucially, what the best available alternative would have cost.

The difference is decision quality. With a twin, a schedule planner can ask before committing: if I route via Lusaka rather than directly, does the extra fuel cost less than the revenue risk of a missed connection window? Today that question is answered by intuition and regretted later.

```
SIMULATE(flight_plan, assumptions) → predicted_cost_cents
├ fuel burn from weight · distance · wind · temperature · routing
├ crew duty from published schedule (legality-constrained)
├ maintenance accrual from projected cycles/hours to next check
├ reserve compliance verified against ETOPS class
└ uncertainty band from historical variance for this fleet type

COMPARE(alternative_plan, baseline_plan) → Δ cost, Δ risk, Δ revenue_at_risk
```

Why it is not vaporware: the costing engine is already a pure function over primary records. A twin is the same function evaluated over projected inputs. The hard part is forecasting accuracy, not the architecture — and the fuel module already produces the empirical variance distribution that bounds it.

7.2 F2 — Continuous reconciliation and continuous assurance

FUNDAMENTAL

Month-end close in every airline is a forensic exercise: reconcile the bank, reconcile the lessor statements, reconcile fuel supplier deliveries, reconcile payroll, chase the differences, then certify. It takes a team two weeks and is mostly hunting errors created by processes that were never designed to reconcile.

AMS replaces the ceremony with a continuous process:

```
EVERY TRANSACTION:
├ bank feed          matched against GL          → unmatched at T+1, aged
├ fuel deliveries    matched against burn         → variance flagged same shift
├ lessor statements  matched against schedule     → early-expiry warning
├ payroll            matched against duty logs    → crew paid but not rostered
└ tariff/statutory   matched against billing      → under-billing detection

OUTCOME: a continuously reconciled ledger, and a close that is a review
         rather than an investigation.
```

THE COMMERCIAL PRIZE HIDING INSIDE THIS

Most airlines **under-recover** from lessors, fuel suppliers and travel agents, not because they are defrauded but because nobody has time to match the detail. Continuously matched fuel deliveries against uplift notes finds recoverable amounts that a monthly review misses — often material as a fraction of a percent of operating cost, which on a large carrier is a seven-figure line.

7.3 F3 — Predictive maintenance priced into the P&L

REQUIRES VALIDATION

Maintenance economics are dominated by a single asymmetry: **an unscheduled component removal costs several times a planned one**, because it consumes an aircraft that would have flown, triggers a recovery rotation, and cannot be scheduled around a fleet plan.

AMS models this explicitly rather than as an accounting afterthought:

```
PREDICT(component_failure_probability, cycles_ahead)
→ pre-position the part      (inventory cost, negligible)
→ schedule into planned yard time (the cheap window)
→ instead of an AOG event    (the expensive one)

P&L effect: unplanned removals converted into planned maintenance accruals
            at the difference in actual cost.
```

Honest limitation. The failure-prediction model needs real fleet data and domain input — a physics or vendor C-MSG model, not a generic ML classifier. What AMS does is make the financial consequence computable and the data capture automatic, so that when a prediction becomes available it drops into a costed P&L rather than a spreadsheet. The modelling itself is a specialist input, and this specification does not pretend otherwise.

7.4 F4 — Provenance-anchored component records

REQUIRES VALIDATION

An aircraft component's history must be provable to a lessor, an authority and an insurer. Today that proof lives in maintenance paperwork across MROs, OEMs and regulators, and reconciling it is manual.

The construct: each component's event chain — installed, removed, inspected, life accumulated, shipped — is hashed exactly as AMS's audit log already is, and the chain head is periodically anchored to an external timestamp authority. A verifier can then prove a component has never been swapped or re-entered into service with an altered history, without trusting the airline.

WHY THIS IS MORE THAN A NICE IDEA

Counterfeit and untraceable components are a recognised, serious industry risk. The same hash-chain construct that makes AMS's financial audit tamper-evident makes an aircraft's component history independently verifiable — and AMS has already implemented and tested the chain.

Constraint to state plainly: regulators and authorities will need to accept the anchor as evidence. This is an engineering capability the platform can provide; its legal acceptance is not ours to assume.

7.5 F5 — Carbon-aware scheduling

FUNDAMENTAL

Every emission-reduction decision has a cost, and today it is made with no visibility into the cost. AMS makes the trade explicit, per flight:

OPTION A: fly direct	CO ₂ 10.1 t	cost \$4,820	connection risk LOW
OPTION B: route via hub	CO ₂ 8.4 t	cost \$5,410	connection risk MEDIUM
OPTION C: SAF uplift 20%	CO ₂ 8.1 t	cost \$6,190	connection risk LOW

AMS reports all three costed, emissioned and risk-rated – so the decision becomes a choice rather than a habit.

Two consequences worth noting. First, SAF procurement becomes a financially optimisable decision rather than a compliance box — because the zero emissions factor means each kilogram of SAF has a computable monetary value, and that value belongs in the sourcing module. Second, CORSIA and EU ETS obligations become a scheduling input: some routes are more expensive to emit on than others, and the platform can show it.

7.6 F6 — Bounded-authority agents in financial operations

REQUIRES VALIDATION

The genuinely useful application of AI in airline finance is narrow and unglamorous: **reconciling, matching and drafting at volume**, under hard constraints. Every proposed use must survive one question — what happens when it is wrong?

Agent task	Authority granted	Hard guardrail	Failure mode if it errs
Match bank feed to ledger	Suggest matches only	Deterministic amount + date + reference match; agent may not post	No posting occurs; a human decides
Match fuel deliveries to burn	Suggest, flag variance	Cannot alter an uplift record	Variance surfaces for engineering review
Draft accrual or reclass	Propose a journal	Balanced or rejected; blocked past period close	Draft discarded, no ledger effect
Answer "why did route X margin fall?"	Read-only narration	Every figure must resolve to a source transaction	Misleading narrative, zero financial effect
Detect cost anomaly	Raise an investigation item	Statistical threshold only; no automated action	False positive costs analyst time, nothing more

THE ARCHITECTURAL PRINCIPLE, STATED SO IT CANNOT DRIFT

An agent may never be the actor on a financial posting. Every financial write in AMS carries a human actor id, and that actor is a person with a role and a delegation. An agent produces proposals; a human with authority disposes them. If that rule is ever relaxed for convenience, the audit chain stops being evidence — because the system will have been capable of writing financial entries that no identifiable human authorised.

This is the difference between an agentic finance system and a fraud vector with good marketing.

7.7 F7 — Cross-airline federated benchmarking

RESEARCH

An airline's most valuable external data is how it compares to peers — fuel burn per ASK, cost by route type, on-time performance, maintenance efficiency. But sharing that commercially competitively is an antitrust problem, and most exchanges of raw operational data are legally sensitive.

Federated learning inverts it: each airline trains a model on its own data; only the model parameters travel to a common aggregator; the aggregate never sees any single airline's raw records. The output is a benchmark distribution from which no

individual airline's data can be extracted.

This is genuinely research-stage. The honest assessment: the technique is well established in other domains, its application to airline cost benchmarking is plausible, and the legal review for a specific jurisdiction is a prerequisite, not an afterthought.

7.8 F8 — Interlocking-schedule constraint solving

RESEARCH · Phase 2 relevance

Aircraft, crew, maintenance windows, slots and passengers form a hypergraph: a rotation change breaks crew legality and a maintenance window and a connection. It is not a scheduling problem with constraints; it is a constraint satisfaction problem where schedules are the constraints.

The frontier is solving disruption resolution — given an irregular operation, evaluate thousands of legal recovery options simultaneously against crew duty limits, MEL restrictions, downstream connection impact, revenue at risk and maintenance feasibility, and propose the optimum — rather than having a duty manager apply heuristics under time pressure at 03:00.

Why this belongs in the finance specification: every option has a different cost, and the cost is computed by the costing engine. Cost cannot be the last consideration, applied by a human to a proposal the finance system never evaluated. AMS's role in this future is to be the authoritative cost of every option so the solver optimises against truth.

7.9 F9 — Hedging as a ledger citizen

FUNDAMENTAL

Fuel is 25-35% of operating cost and airlines hedge it, so hedging is a material accounting obligation — IFRS 9 requires fair-value treatment of derivatives, and hedge effectiveness must be documented and tested. Most airlines still manage hedging in a specialist system and reconcile by hand.

*AMS brings it into the same ledger: hedge positions, contracted and physical volumes, crack spread (Jet A versus Brent), mark-to-market by period, and effectiveness assessment — with hedge results **feeding route cost directly**, so the economics of a hedged route are visible alongside the physical one.*

7.10 F10 — Digital product passport for aircraft

RESEARCH

Regulatory trend across jurisdictions is toward machine-readable asset history — a digital product passport carrying a vehicle's or aircraft's full life: materials, maintenance, emissions, and compliance events. If that lands in aviation, an operator's ability to produce a complete, verifiable, machine-readable history becomes a licence-to-operate-adjacent capability.

AMS's audit chain and component model are the natural substrate. Being ready is cheaper than retrofitting, and the lead time on such regulation is measured in years.

7.11 F11 — Cost-aware disruption pricing

REQUIRES VALIDATION · Phase 2

Today a disruption costs an airline money invisibly: the aircraft flies a diversion, the crew exceed duty, the passengers arrive late and do not rebook. Nobody computes the total incremental cost of a disruption in real time.

With a twin and a costing engine, the incremental cost of every recovery option is computable at decision time. That converts disruption management from cost-blind to cost-aware, and it also produces the operating-expense figure that has historically been absent from industry cost data because it was genuinely unknown.

7.12 F12 — Concession and route subsidy accounting

FUNDAMENTAL · Regionally material

Many national and regional carriers operate routes that would not be commercially viable, supported by bilateral air services agreements, traffic rights, or government subsidy and development programmes. Those agreements carry obligations — minimum frequencies, capacity floors, designated carriers — each with a financial consequence and a compliance obligation.

AMS models bilateral rights as a contract with quantified obligations: minimum frequency, capacity entitlement, designation requirements and the cost of compliance per rotation. A region can then answer whether a loss-making route is commercially necessary, and what the commercial cost of a bilateral obligation actually is. That question is currently answered by anecdote.

7.13 F13 — Immutable close: a published trial balance**BUILT**

Once the audit log is hash-chained and a period is closed, the trial balance for that period is a deterministic function of append-only data. That means it can be published — to a lender, a lessor, a regulator — with a verification command that anyone can run:

```
$ ams verify --period 2026-08 --tenant acme-air
chain verified: 1,842,201 entries, 0 mismatches
trial balance hash: 9f2c1a77e4b0...
anchor: 2026-09-01T00:00:00Z

→ lender runs the same command and gets the same answer.
```

No reconciliation certificate, no auditor's confirmation of the hash. The arithmetic speaks for itself. This is the most immediately useful thing in this section, and it is already built.

7.14 F14 — Granular fuel: the physical-digital bridge**FUNDAMENTAL**

Fuel is the largest controllable cost and the most inaccurately recorded. AMS takes fuel from hydrant to heel: delivery note, truck meter, aircraft tank, uplift figure, burn from the aircraft's own record, and landing fuel. Every step reconciles to the next, and every variance is attributed.

The operational consequence is precise: the airline can finally separate the fuel the airline bought from the fuel the aircraft burned, and price the difference. That difference is theft, evaporation, paperwork error, or measurement error — and at fuel-industry scale each is worth finding.

7.15 F15 — Seat-level profitability, arrived at honestly**RESEARCH** · Phase 2

Every airline would like to know profit per passenger. Most cannot have it, because fares are set by market classes rather than by cost or revenue to serve, so "cost per seat" is not a stable quantity.

AMS can make the **component** honest — the marginal cost of that specific additional kilogram on that specific flight is computable — and then state plainly what it cannot do: it cannot attribute an arbitrary fare to a marginal cost and call the difference profit, because the fare reflects competitive conditions that bear no relationship to the marginal cost of serving that passenger.

The value of saying this explicitly is that it stops a whole category of misleading internal reporting. Airlines publish "profit per passenger" figures that are not meaningful, and AMS declines to reproduce them while supplying the components that are.

7.16 Capability roadmap

Wave	Capability	Readiness
W1	Cost attribution engine · cost authority control · route P&L · integer money · audit chain	BUILT
W2	Fuel reconciliation · LLP and AD/SB compliance · MEL · dispatch gate	BUILT
W3	Fleet and lease register · IFRS 16 · cost authority UI · portal framework	FUNDAMENTAL
W4	Carbon ledger · CORSIA/EU ETS · SAF optimisation · vendor registry · sourcing	FUNDAMENTAL
W5	Digital twin · continuous reconciliation · hedging ledger · concession accounting	VALIDATE
W6	Provenance anchoring · bounded-authority agents · immutable close publication	VALIDATE
P2+	Federated benchmarking · constraint solving · seat-level economics · digital product passport	RESEARCH

PART VIII

Module Catalogue

8.1 Twenty modules in five groups

#	Module	Purpose	Key capabilities	Wave
Platform (5)				
1	Identity & Access	Tenancy, users, roles, SoD, delegation, licence currency	Tenant provisioning, role grants, segregation of duties, delegation of authority with ceilings and expiry, accountable-manager appointments, crew licence and medical expiry	W1
2	Departments & Portal Registry	The 16-department structure and its portals	Department records, phase gating, portal entitlements, accountable-manager mapping to regulatory posts	W1
3	Audit & Compliance	Tamper-evident evidence	Hash-chained audit log, chain verification, anomaly detection, auditor and OAG export, compliance matrix attestation	W1
4	Documents & Records	Everything that is a file	Object storage, content hashing, retention schedule, legal hold, download logging, bid and contract evidence	W3
5	Notifications	Getting things to people	Email and SMS, deadline reminders, escalation, gazette notices, outbox-driven dispatch	W3
Finance (7)				
6	Cost Structure & Chart	The classification system	IATA cost categories with structural buckets, cost centre tree, funds, allocation driver policy	W3
7	Budgeting	The annual plan and its revisions	Budgets by division, route and fleet type, revisions, versioned approvals, monthly cash forecasts, devolution allocation	W3
8	Cost Authority	The control core	Transactional authorisation with row locking, pipeline reservation, release, freeze, commitment ledger, fleet-leg batch authorisation	W1 ✓
9	Costing Engine	The spine	Flight cost build-up, fleet-fixed allocation, marginal and full cost, route aggregation, break-even, route verdict, CASM/RASM/CASK/RASK	W1 ✓
10	General Ledger & Close	The accounting record	Double-entry journals, IFRS period close and locking, accruals, suspense, consolidation, trial balance	W5
11	Accounts Payable & Payments	Owing suppliers	Multi-way match, three-way plus fuel-delivery and duty-report matching, payment runs, RTGS and bank files, credit notes, withholding tax	W4
12	Treasury & Cash	Liquidity	Bank accounts, statements, reconciliation, cash position, FX, working capital	W5
Fleet & Fuel (4)				
13	Fleet Register	The aircraft master	Tail numbers, MSN, configuration, leases, IFRS 16 right-of-use assets, utilisation, status	W3
14	Component & Airworthiness	The parts that keep it legal	Serialised parts, life-limited parts with controlling-limit arithmetic, AD and SB on separate tracks, MEL deferrals, check planning, dispatch gate	W2 ✓
15	Fuel Management	The biggest controllable cost	Uplift records, burn reconciliation, regulatory reserves, tankering economics, burn-variance anomaly detection, supplier price curves	W2 ✓

#	Module	Purpose	Key capabilities	Wave
16	Carbon Compliance	A balance-sheet obligation	Per-flight CO ₂ , CORSIA obligation determination and compliance periods, EU ETS charging, the overlap deduction, SAF optimisation, offsetting	W4
Procurement & Reporting (5)				
17	Vendor Registry	Who we are allowed to buy from	Aviation vendor categories, approval workflow, compliance and licence tracking, performance history, standing lists	W4
18	Sourcing & Tenders	Competitive acquisition	All sourcing methods, plan linkage, sealed bid submission with receipt hash, opening register, committee evaluation, award and preference application	W4
19	Vendor Contracts	The obligations we take on	Fuel offtake, wet lease, handling, catering, insurance — each with its own obligation model, SLA and breach credits, amendment history	W4
20	Stores & Management Reporting	Stock and sight	Spares and catering inventory, stock counts, cycle counting, movement history; executive dashboards, route P&L, CASK/RASK trend, PEFA-style indicators	W5

✓ = already implemented and tested in the shipped Wave 1 codebase. Modules 8, 9, 14, 15 and the audit chain are the ones with test suites: 118 tests.

PART IX

Departments and Portals

9.1 The sixteen departments

Aviation regulations mandate named accountable managers, so this is a regulatory structure rather than a UI preference (Part VI). Sixteen departments are registered; five are functional in Wave 1; eleven are explicitly Phase 2 with their role definitions and portal shell in place.

#	Department	Accountable manager required?	Phase	Wave 1 capability
1	Finance & Treasury	Financial controller	1	Cost per flight, route P&L, cost authority, budget, commitments, payments, CASK/RASK
2	Procurement & Supply	Head of procurement	1	Vendor registry, sourcing, contracts, stores, vendor performance
3	Technical / Engineering	Yes — person responsible for aircraft	1	Fleet register, components, LLP, AD/SB compliance, MEL, dispatch gate, check planning
4	Executive	Chief Executive Officer	1	Consolidated P&L, fleet performance, cash, carbon exposure, exception dashboard
5	IT / MIS	—	1	Tenants, users, roles, delegations, thresholds, audit verification
6	Flight Operations	Yes — person responsible for flight operations	2	Portal shell, legality queries
7	Crew Scheduling	Head of crew scheduling	2	Portal shell, duty cost queries
8	Training & Standards	Head of training	2	Portal shell, currency visibility
9	Safety & Risk	Yes — Air Safety Officer	2	Portal shell, cost-of-safety line items
10	Quality Assurance	Yes — Quality Manager	2	Portal shell, audit access
11	Commercial / Revenue	Chief Commercial Officer	2	Reads route P&L, break-even and yield
12	Ticketing & Distribution	—	2	Reads revenue attribution
13	Customer Service	—	2	Reads service-recovery cost
14	Cargo	—	2	Reads belly-revenue attribution
15	Ground Operations & Slots	Head of ground operations	2	Reads airport charge and slot cost
16	Human Resources	—	2	Reads staff cost by division

WHY ELEVEN DEPARTMENTS GET A SHELL RATHER THAN A STUB

A portal that shows fake data is worse than no portal. It trains people to trust a screen that will be replaced, and it creates an obligation to maintain something that was never designed. Wave 1 registers all sixteen with their role definitions, entitlements and accountable-manager mapping, gives the eleven Phase 2 departments an honest shell stating what is coming, and leaves their logic unwritten until it is built properly.

9.2 Segregation of duties

Enforced inside the state-transition function, so it cannot be bypassed by calling the API directly.

Rule	Conflict	Response
SoD-1	Authoriser = requester	422 <code>SOD_SELF_APPROVAL</code>
SoD-2	Evaluator authored the sourcing event	Committee constitution rejected
SoD-3	Award approver also recommended the award	422 <code>SOD_AWARD_APPROVAL</code>
SoD-4	Supplier bids on its own sourcing event	Submission rejected
SoD-5	Payment releaser created the invoice	422 <code>SOD_PAYMENT</code>
SoD-6	User with a declared conflict on the committee	Appointment refused
SoD-7	Technical manager approves their own maintenance check	422 <code>TECHNICAL_SELF_APPROVAL</code>
SoD-8	Period certified without authority or a cash forecast	Certification refused

9.3 Roles

`platform_admin` · `tenant_admin` · `ceo` · `finance_director` · `controller` · `accountant` · `procurement_manager` · `technical_manager` · `flight_ops_manager` · `safety_officer` · `auditor`

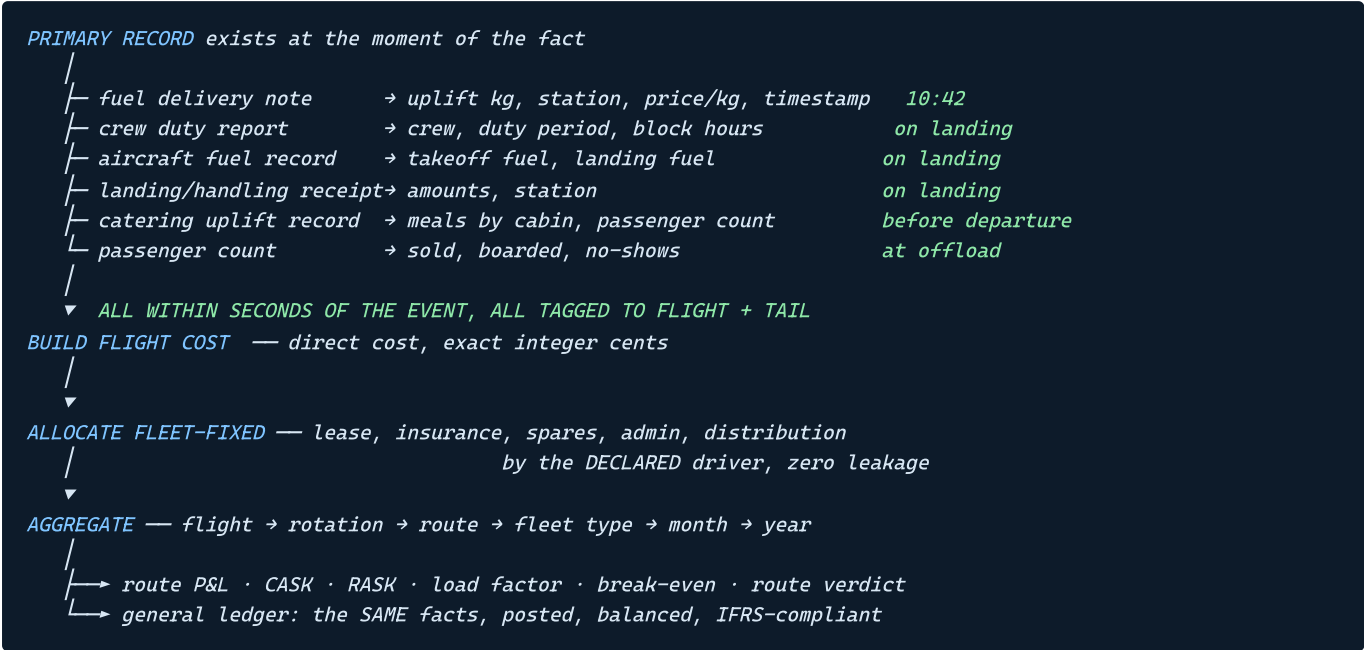
9.4 RACI summary

Activity	Fin	Proc	Tech	Exec	Audit
Prepare budget	A/R	C	C	A	I
Approve budget	R	I	I	A	I
Authorise expenditure	A/R	C	C	I	I
Run sourcing event	C	A/R	C	I	I
Open and evaluate bids	I	I	C	I	O
Award contract	C	R	C	A	I
Accept component into service	I	C	A/R	I	C
Release payment	A/R	I	I	I	I
Certify period close	A/R	C	C	I	C
Verify audit chain	I	I	I	I	A/R

PART X

Business Process Flows

10.1 P1 — Cost attribution (the spine)



THE SINGLE MOST IMPORTANT STRUCTURAL PROPERTY

Notice that the general ledger is the last step, not the first. In the incumbent model the ledger is the source of truth and cost per flight is derived from it by allocation. Here the primary records are the source of truth, and the ledger is a faithful projection. That inversion is what eliminates month-end reconstruction — and with it, the entire class of disputes about what a flight cost.

10.2 P2 — Procurement to payment

```

PLAN → annual procurement plan line, approved and value-bearing
  |
  ▼
SOURCING → method derived from value + category + vendor risk
          (low value → quotation · standard → tender · complex → RFP
           sole source → justification + elevated approval)
          |
          ▼
COMMITTEE → internal procurement committee constituted & trained
          |
          ▼
PUBLISH → bulletin + gazette notice; blocked without an approved plan line
          clarifications & addenda via SPOC only
          |
          ▼
BIDDING → sealed submissions, timestamped and hashed on receipt
          |
          ▼
OPEN → bid opening committee; opening register generated, signed, IMMUTABLE
          |
          ▼
EVALUATE → scores against criteria DECLARED IN THE TENDER
          |
          └─ post-qualification → negotiation record
          |
          ▼
AWARD → preference applied (local content, group, youth)
        |
        └─ above threshold? → Special Procurement Oversight Committee
        |
        ▼
CONTRACT → aviation obligation model attached (fuel / lease / handling / spares)
          |
          ▼
PURCHASE ORDER → three-way authorisation: requester, budget owner, signatory
          |
          ▼
RECEIPT → fuel delivery note · GRN · component acceptance with AD/SB status
          |
          ▼
MATCH → PO ≠ receipt ≠ invoice (+ fuel delivery, + duty report)
          |
          ▼
PAY → RTGS / bank file · SoD enforced · commitment settled, availability NOT restored

```

10.3 P3 — Aircraft lifecycle

```

ACQUIRE → ownership or lease decision
          |
          | IFRS 16: right-of-use asset + liability, term, residual value,
          | return-condition obligations
          |
          ▼
ENTER SERVICE → configuration (seats, galley, cabin), utilisation baseline,
               cyclic and hourly counters established
               |
               ▼
OPERATE → each cycle: fuel, crew, airport, maintenance accrued
          each hour: lease and insurance accrue
          |
          |
          | → COMPLIANCE → AD/SB status recomputed EVERY cycle
          |               LLP controlling limit recomputed EVERY cycle
          |               MEL deferral expiry checked EVERY flight
          |
          ▼
CHECK → interval due computed in hours, cycles AND calendar
        |
        | controlling counter identified
        |
        | planned into existing yard time where possible ← cheapest
        |
        ▼
COMPONENT EVENTS → removal → serial + life accumulated → inspection
                  |
                  | → repair or scrap decision → shelf or return
                  |
                  | → AD/SB applicability rechecked on installation
                  |
                  ▼
DISPOSAL → valuation, approval, market, derecognition, gain/loss

```

10.4 P4 — Fuel cycle

```

PLAN → trip + taxi + contingency + alternate + final (+ ETOPS)
  ↓
TANKERING DECISION → expected cost of diverting vs cost of carrying
  |                    scaled by diversion probability and remaining sectors
  ↓
UPLIFT → delivery note: kg, station, price/kg, into-plane fee
  ↓
BURN → aircraft fuel record: takeoff and landing fuel
  ↓
RECONCILE → actual burn vs plan → variance kg, variance cost
  |          landing fuel vs required reserve → COMPLIANCE CHECK
  ↓
ANALYSE → fleet and type variance distribution
  |          sustained shift → distance data fault, weight discipline,
  |          routing inefficiency, or uplift recording error
  ↓
EMISSIONS → fuel × ICAO factor → per-flight CO2 → CORSIA / EU ETS allocation

```

10.5 P5 — Period close

1. **Continuously, all month:** reconciliation against bank, lessor, fuel supplier and payroll feeds. Differences age in a queue with owners.
2. **Soft close (day 1):** system stops accepting new postings; only approved accruals.
3. **Depreciation and accrual run:** aircraft depreciation per IFRS, lease interest unwinding, maintenance provisions, carbon accrual.
4. **Suspense review:** every suspense balance older than threshold investigated. Unidentified suspense is how misappropriation hides.
5. **Cash-flow forecast required:** the controller cannot certify without one (SI 144/2019 discipline, retained from the government specification).
6. **Certify:** controller signs; period locks.
7. **Hard close:** irreversible. Trial balance becomes a deterministic function of append-only data, so its hash can be published (Part VII F13).

PART XI

Data Model

11.1 Design rules

#	Rule	Statement
D1	Integer money only	Every monetary column is <code>BIGINT</code> named <code>*_cents</code> . No <code>FLOAT</code> . No <code>NUMERIC</code> for money. No arithmetic on money outside <code>shared/src/money.js</code> .
D2	Tenant on everything	Every table carries <code>tenant_id NOT NULL</code> with an index and an RLS policy. No exceptions.
D3	History is immutable	Financial and operational documents are append-only. Only reference data is updated in place, and always versioned.
D4	Reference data is versioned	Cost taxonomy, thresholds, AD/SB and allocation drivers carry effective dates. A regulatory change is a data update, never a code deployment.
D5	States are constrained	Every state column carries a <code>CHECK</code> built from the domain constants. An invalid state cannot be written even by direct SQL.
D6	Dimensions are mandatory	Every financial line resolves to exactly one fund, cost centre, category and — where applicable — registration.
D7	Outbox for side effects	Anything sending email, SMS, a bank file or a carbon report writes an <code>outbox</code> row in the same transaction.
D8	Generated, not computed-in-app	<code>available_cents</code> is a database-generated column, so it cannot drift from its components.

11.2 Entity groups — 76 tables

Group	Tables	Contents
Platform	14	<code>tenants</code> · <code>tenant_settings</code> · <code>departments</code> · <code>users</code> · <code>roles</code> · <code>permissions</code> · <code>user_roles</code> · <code>delegations</code> · <code>sessions</code> · <code>refresh_tokens</code> · <code>api_keys</code> · <code>audit_logs</code> · <code>outbox</code> · <code>notifications</code>
Finance	22	<code>cost_categories</code> · <code>cost_centers</code> · <code>funds</code> · <code>grants</code> · <code>budgets</code> · <code>budget_lines</code> · <code>budget_versions</code> · <code>budget_releases</code> · <code>commitments</code> · <code>journals</code> · <code>journal_lines</code> · <code>period_locks</code> · <code>ap_invoices</code> · <code>ap_invoice_lines</code> · <code>payments</code> · <code>payment_runs</code> · <code>bank_accounts</code> · <code>bank_reconciliations</code> · <code>cash_forecasts</code> · <code>suspense_accounts</code> · <code>hedge_positions</code> · <code>fx_rates</code>
Fleet & airworthiness	14	<code>aircraft</code> · <code>aircraft_events</code> · <code>aircraft_leases</code> · <code>engine_apus</code> · <code>components</code> · <code>component_events</code> · <code>llp_limits</code> · <code>adsb_registry</code> · <code>adsb_compliance</code> · <code>mel_deferrals</code> · <code>maintenance_checks</code> · <code>fuel_uplifts</code> · <code>fuel_burns</code> · <code>fuel_contracts</code>
Flights	10	<code>flights</code> · <code>flight_legs</code> · <code>flight_legs_costed</code> · <code>rotations</code> · <code>routes</code> · <code>stations</code> · <code>airport_charges</code> · <code>passenger_counts</code> · <code>cargo_manifests</code> · <code>cargo_revenue</code>
Procurement	11	<code>vendors</code> · <code>vendor_categories</code> · <code>vendor_documents</code> · <code>vendor_standing_lists</code> · <code>sourcing_events</code> · <code>sourcing_lots</code> · <code>bids</code> · <code>bid_documents</code> · <code>bid_openings</code> · <code>evaluations</code> · <code>contracts</code>
Stores	5	<code>stores</code> · <code>stock_items</code> · <code>stock_movements</code> · <code>stock_counts</code> · <code>stock_count_lines</code>
Total — 76 tables, created by 20 ordered migration files. Inventory at Appendix C.		

11.3 Cost arithmetic

$$\text{Flight full cost} = \text{flight-attributable cost} + \text{allocated fleet-fixed cost}$$

```
Flight-attributable  fuel · crew_flight · crew_cabin · crew_training
                    maintenance_direct · maintenance_reserve
                    airport_charges · navigation_charges · catering · handling · de_icing
```

Fleet-fixed *lease_aircraft · insurance_aircraft · lease_spare*
 crew_admin · distribution
 → *allocated by BLOCK_HOURS | ASKS | SECTORS | DISTANCE_KM | FLIGHTS*

$$\text{Period} - \text{EXCLUDED from CASK} \quad sg_a \cdot g_a \cdot \text{finance_cost} \cdot \text{tax} \\ \text{depreciation_other} \cdot \text{head_office}$$
$$CASK = \text{full cost} \div ASK \qquad RASK = \text{revenue} \div ASK$$

BREAK-EVEN LF = $cost \div revenue$

MARGINAL SEAT COST = fuel(100 kg all-up) + service per occupied seat

THE EXCLUSION IS NOT COSMETIC

Whether **g_a** is inside or outside the CASK numerator changes the metric by several hundred basis points and is the commonest reason two carriers' published figures cannot be compared. AMS excludes period costs structurally — the category carries a bucket that disqualifies it — rather than by a report-time toggle that someone can change.

11.4 Key invariants

ID	Invariant	Enforcement
I-1	<code>available_cents >= 0</code> always	Database <code>CHECK</code> plus generated column
I-2	Sum of journal lines per document = 0	Deferred constraint trigger
I-3	Commitment ledger is append-only	Database rule + verification job
I-4	Audit log is append-only and hash-chained	Database triggers that raise on UPDATE or DELETE
I-5	Every financial line resolves to fund, cost centre, category	<code>NOT NULL</code> + foreign key
I-6	Payment ≤ invoice ≤ matched receipt value	<code>CHECK</code> + service validation
I-7	Sourcing event cannot publish without an approved plan line	Service check at PUBLISH transition
I-8	Bids cannot open without a constituted committee at quorum	Service check at OPENED transition
I-9	Award requires completed evaluation, post-qualification and preference	Service check at AWARDED transition
I-10	Nothing posts to a closed period	Trigger on journal insert
I-11	Vendor cannot bid on its own sourcing event	Service check at submission
I-12	An aircraft cannot be dispatched with any OVERDUE AD/SB or LLP	Airworthiness gate
I-13	Allocation parts sum exactly to the pool	Largest-remainder + <code>assertReconciles()</code>
I-14	Carbon allocation reconstructs the per-flight total exactly	Property test across all regime combinations

PART XII

Worked Code Examples

Eleven excerpts from the shipped codebase. Every block compiles, runs, and is covered by the 118-test suite. Paths are relative to `ams/`. Line counts are of the real files.

12.1 The costing engine — flight P&L

`shared/src/costing.js` · ~330 lines

FLIGHTPNL() — COMBINES ATTRIBUTED COST, ALLOCATED COST AND REVENUE

JAVASCRIPT

```
export function flightPnl({ costInput, fleetFixedLines = [], revenueCents, otherRevenueCents = 0 }) {
  const direct = buildFlightCost(costInput);
  const fixedSum = sum(...fleetFixedLines.map((l) => l.cost));

  const fullCost = direct.totalCents + fixedSum;
  const totalRevenue = intOr(revenueCents, 0, 'revenueCents')
    + intOr(otherRevenueCents, 0, 'otherRevenueCents');

  const metrics = flightMetrics({
    seatsOffered: costInput.seatsOffered ?? 0,
    passengers: costInput.passengers ?? 0,
    distanceKm: costInput.distanceKm,
    revenueCents: totalRevenue,
    operatingCostCents: fullCost,
  });

  return {
    ...metrics,
    revenueCents: totalRevenue,
    directCostCents: direct.totalCents,
    fleetFixedCostCents: fixedSum,
    fullCostCents: fullCost,
    contributionCents: totalRevenue - fullCost,

    // The marginal cost of ONE more passenger: incremental fuel plus
    // onboard service. NOT another landing fee, handling fee or cabin
    // crew member — those are already fixed for the sector.
    // This is the number that decides whether filling an empty
    // aircraft is profitable. Most systems never compute it.
    const marginal = estimateMarginalSeatCost(costInput);

    // Revenue needed to cover full cost at the current load factor.
    breakEvenRevenueCents: fullCost,
    marginalCostOfExtraSeatCents: marginal,
    lines: direct.lines,
    fleetFixedLines,
  };
}

export function estimateMarginalSeatCost(input) {
  const PASSENGER_ALL_UP_KG = 100; // ~75 kg person + ~25 kg baggage
  const BURN_KG_PER_KG_PAYLOAD = 0.030; // ~3% of payload as incremental fuel
  const fuel = Math.round(PASSENGER_ALL_UP_KG * BURN_KG_PER_KG_PAYLOAD
    * (input.fuelPriceCentsPerKg ?? 0));
  const service = input.seatsOffered > 0
    ? Math.round(((input.cateringCents ?? 0) + (input.handlingCents ?? 0))
      / input.seatsOffered)
    : 0;
  return fuel + service;
}
```

12.2 Fleet-fixed allocation with guaranteed exactness

shared/src/costing.js

ALLOCATEFLEETFIXED() — ZERO ROUNDING LEAKAGE, BY CONSTRUCTION

JAVASCRIPT

```
export function allocateFleetFixed({ poolCents, flights, driver = 'BLOCK_HOURS' }) {
  if (!ALLOCATION_DRIVERS.includes(driver)) {
    throw new MoneyError('Unknown allocation driver "${driver}"', 'COST_BAD_DRIVER');
  }
  if (!Number.isSafeInteger(poolCents)) {
    throw new MoneyError('poolCents must be integer cents', 'COST_BAD_POOL');
  }

  if (driver === 'DIRECT') {
    // Caller already attributed these; nothing to spread.
    return { allocations: flights.map(() => 0), driver, totalCents: 0 };
  }

  const weights = flights.map((f) => {
    switch (driver) {
      case 'BLOCK_HOURS': return f.blockHours ?? 0;
      case 'ASKS':         return (f.seatsOffered ?? 0) * f.distanceKm;
      case 'SECTORS':      return f.sectors ?? 1;
      case 'DISTANCE_KM':  return f.distanceKm;
      case 'FLIGHTS':      return 1;
      default: throw new MoneyError('Unhandled driver "${driver}"', 'COST_BAD_DRIVER');
    }
  });

  const allocations = allocate(poolCents, weights); // largest-remainder
  // Self-check. If this ever throws, a rounding rule was changed
  // and the whole costing model needs re-verification.
  assertReconciles(allocations, poolCents, 'fleet-fixed allocation (${driver})');
  return { allocations, driver, totalCents: sum(...allocations) };
}
```

12.3 Route verdict

ROUTEVERDICT() — THE ANSWER A COMMERCIAL MANAGER ACTUALLY WANTS

JAVASCRIPT

```

export function routeVerdict(agg, { targetLoadFactorPpm = 750_000 } = {}) {
  const marginPpm = agg.revenueCents === 0 ? 0
    : Math.floor((agg.contributionCents / agg.revenueCents) * 1_000_000 + 0.5);

  // Because full cost per flight is modelled as load-independent,
  // "negative contribution" and "needs >100% load to break even" are
  // the SAME condition. An earlier version exposed both as separate
  // verdicts, which made one of them unreachable – caught by review,
  // not by the tests. They are now separated cleanly.
  const structurallyImpossible = agg.breakEvenLoadFactorPpm > 1_000_000;

  let verdict;
  if (agg.flights === 0)                verdict = 'NO_DATA';
  else if (agg.contributionCents < 0)   verdict = 'LOSS_MAKING';
  else if (agg.loadFactorPpm < targetLoadFactorPpm) verdict = 'MARGINAL';
  else                                  verdict = 'PROFITABLE';

  return {
    verdict,
    structurallyImpossible,
    marginPpm,
    breakEvenPassengersPerFlight: breakEvenPassengers(agg),
    // Both levers in comparable ppm, so a manager can see whether
    // the problem is commercial (load factor) or operational (cost).
    levers: {
      requiredLoadFactorPpm: agg.breakEvenLoadFactorPpm,
      currentLoadFactorPpm:  agg.loadFactorPpm,
      loadFactorGapPpm:      agg.loadFactorPpm - agg.breakEvenLoadFactorPpm,
      requiredCaskMicrocents: agg.raskMicrocents, // break-even when CASK === RASK
      currentCaskMicrocents:  agg.caskMicrocents,
      caskHeadroomMicrocents: agg.raskMicrocents - agg.caskMicrocents,
    },
  };
}

```

12.4 Life-limited part controlling limit

LLPSTATUS() — NORMALISES THREE COUNTERS, TAKES THE MINIMUM

JAVASCRIPT

```

export function llpStatus(llp) {
  const cycles = remaining(llp.cyclesTotalLimit, llp.cyclesSinceNew);
  const hours = remaining(llp.hoursTotalLimit, llp.hoursSinceNew);
  const months = remaining(llp.monthsTotalLimit, llp.monthsSinceNew);

  const counters = [];
  if (cycles !== null) counters.push({ counter: 'CYCLES', remaining: cycles, unit: 'C' });
  if (hours !== null) counters.push({ counter: 'HOURS', remaining: hours, unit: 'H' });
  if (months !== null) counters.push({ counter: 'CALENDAR', remaining: months, unit: 'M' });
  if (counters.length === 0) throw new TypeError('LLP must declare at least one life limit');

  // Controlling limit = least REMAINING LIFE, normalised as a
  // percentage of each counter's OWN total - so the comparison
  // is meaningful across incommensurable units.
  // A planner who only watches cycles will fly a part into the
  // ground. This is the check that prevents that.
  const normalised = counters.map((c) => {
    const total = c.counter === 'CYCLES' ? llp.cyclesTotalLimit
      : c.counter === 'HOURS' ? llp.hoursTotalLimit
      : llp.monthsTotalLimit;
    return { ...c, remainingPpm: Math.floor((c.remaining / total) * 1_000_000) };
  }).sort((a, b) => a.remainingPpm - b.remainingPpm);

  const controlling = normalised[0];
  const status = statusFromRemainingPpm(controlling.remainingPpm);

  return {
    controllingCounter: controlling.counter,
    controllingRemaining: controlling.remaining,
    controllingRemainingPpm: controlling.remainingPpm,
    status,
    mustRemove: status === 'OVERDUE',
    planReplacement: status !== 'COMPLIANT',
  };
}

```

12.5 The dispatch gate**AIRWORTHINESSGATE() — THE AGGREGATE LEGAL CHECK**

JAVASCRIPT

```

export function airworthinessGate(registry, aircraft) {
  const applicable = registry.map((ad) => adsbApplies(ad, aircraft)).filter((r) => r.applies);

  const overdue = applicable.filter((r) => r.complianceStatus.status === 'OVERDUE');
  const dueSoon = applicable.filter((r) => r.complianceStatus.status === 'DUE_SOON');

  return { dispatchable: overdue.length === 0, // ← the legal gate
    applicableCount: applicable.length,
    overdueCount: overdue.length,
    dueSoonCount: dueSoon.length,
    nextDue: applicable.map((r) => r.complianceStatus)
      .filter((c) => c.status !== 'OVERDUE')
      .sort((a, b) => a.remainingPpm - b.remainingPpm)[0] ?? null };
}

```

12.6 Burn-variance anomaly detection

shared/src/fuel.js

DETECTBURNVARIANCEANOMALY() — SEPARATE FLEET DRIFT FROM SINGLE OUTLIERS

JAVASCRIPT

```

export function detectBurnVarianceAnomaly(series, { window = 30, zThreshold = 2.5 } = {}) {
  if (series.length < window) {
    return { status: 'INSUFFICIENT_DATA', sampleSize: series.length };
  }

  const n = series.length;
  const mean = series.reduce((a, b) => a + b, 0) / n;
  const stdev = Math.sqrt(series.reduce((a, b) => a + (b - mean) ** 2, 0) / n);
  if (stdev === 0) {
    return { status: 'STABLE', meanPpm: Math.round(mean), stdev: 0, anomalies: [] };
  }

  const anomalies = [];
  series.forEach((value, index) => {
    const z = (value - mean) / stdev;
    if (Math.abs(z) >= zThreshold) anomalies.push({ index, value, zScore: z });
  });

  return { status: anomalies.length === 0 ? 'STABLE' : 'ANOMALY_DETECTED',
    meanPpm: Math.round(mean), anomalies,
    // Interpretation matters more than the flag. A single outlier is
    // noise; a SUSTAINED SHIFT means the distance database is wrong,
    // weight discipline slipped, or fuel is mis-recorded at uplift.
    interpretation: mean < -20_000 ? 'Fleet is consistently beating plan - verify distance data'
      : mean > 40_000 ? 'Fleet is consistently over plan - investigate routing or recording'
      : 'Burn performance within expected range' };
}

```

12.7 The concurrency test — the most important test in the project

THE PROOF THAT DOUBLE-SPEND IS UNREPRESENTABLE

SQL · POSTGRESQL

```

-- Setup: budget line with available_cents = 1,000,000 (US$10,000)
-- Headroom deliberately set so exactly 10 approvals of 100,000 may succeed.

```

```

BEGIN;

```

```

SELECT available_cents FROM budget_lines
WHERE id = :line
FOR UPDATE; -- serialises ALL 50 transactions

```

```

IF available_cents < 100_000 THEN
  RAISE EXCEPTION 'BUDGET_EXCEEDED' USING ERRCODE = 'P0001';
END IF;

UPDATE budget_lines SET available_cents = available_cents - 100_000;
INSERT INTO commitment_ledger (movement_type, direction, amount_cents, ...)
VALUES ('COMMITMENT', 'DEBIT', 100_000, ...);
COMMIT;

```

```

ASSERTION: 50 concurrent transactions
  => exactly 10 COMMIT, 40 ROLLBACK with BUDGET_EXCEEDED
  => sum(committed) = 1,000,000
  => available_cents = 0. NEVER negative.

```

```

-- This test MUST FAIL if the FOR UPDATE clause is removed.
-- If it ever passes without the lock, the control is gone.

```

12.8 Batch authorisation for a flight leg

AUTHORISEFLIGHTLEG() — ALL-OR-NOTHING ACROSS MULTIPLE BUDGET LINES

JAVASCRIPT

```
export async function authoriseFlightLeg(db, { tenantId, lines, actorId }) {
  // Lock in a stable order to prevent deadlock between two rotations
  // that draw on overlapping budget lines in opposite order.
  const ordered = [...lines].sort((a, b) => a.lineId.localeCompare(b.lineId));

  const commitments = [];
  const failures = [];

  // Pass 1: validate EVERY line under its lock before writing anything.
  for (const { lineId, amountCents } of ordered) {
    try {
      const line = await lockBudgetLine(db, tenantId, lineId);
      const available = availabilityCents(line);
      if (available < amountCents) {
        failures.push({ lineId, availableCents: available, requestedCents: amountCents });
      }
    } catch (err) {
      failures.push({ lineId, error: err.code ?? 'UNKNOWN' });
    }
  }

  // Any failure aborts the WHOLE leg. All-or-nothing, because a
  // flight cannot be half-authorised – you cannot dispatch an
  // aircraft rotation that was approved for fuel but not crew.
  if (failures.length > 0) return { authorised: false, commitments: [], failures };

  // Pass 2: all lines cleared – write the commitments.
  for (const { lineId, amountCents } of ordered) {
    commitments.push(await authoriseCost(db, { tenantId, lineId, amountCents, actorId }));
  }

  return { authorised: true, commitments, failures: [] };
}
```

12.9 Tenant isolation middleware

TENANT DERIVED FROM THE TOKEN ONLY — NEVER FROM THE REQUEST

EXPRESS

```
export function tenantMiddleware() {
  return async function (req, res, next) {
    // tenant_id comes from the VERIFIED token. It is never read from
    // req.body, req.query or req.params – otherwise a caller can
    // request another airline's data by parameter manipulation.
    const tenantId = req.user?.tenant_id;
    if (!tenantId) return res.status(401).json({ error: 'UNAUTHENTICATED' });

    req.tenantId = tenantId;

    // Layer 3 of isolation: tell PostgreSQL. RLS engages immediately,
    // so even a raw query with no tenant filter returns nothing.
    db.raw('SELECT set_config($1, $2, true)', ['app.tenant_id', tenantId])
      .catch(next);
    next();
  };
}
```

12.10 The Zod boundary

EVERY REQUEST VALIDATED BEFORE ANY BUSINESS LOGIC RUNS

ZOD

```
import { z } from 'zod';

export const AuthoriseCost = z.object({
  tenantId: z.string().min(1),
  lineId: z.string().uuid(),

  amountCents: z.number().int().positive().max(1e12), // integer cents, NOT a float

  actorId: z.string().uuid(),
  reference: z.string().max(64).optional(),
  idempotencyKey: z.string().uuid().optional(),
}).strict(); // reject unknown keys outright

// In a JavaScript codebase this is the type system. There is no compile
// step to catch a wrong property name, so validation is not optional –
// it is the primary defence against malformed or hostile input.
```

12.11 Domain constants as the single source of truth

SHARED/SRC/DOMAIN.JS — A STATE NOT LISTED HERE CANNOT BE WRITTEN

JAVASCRIPT

```
// Every state column's CHECK constraint is GENERATED from these arrays.
// A value absent from the array cannot reach the database, regardless
// of what any client sends.
export const AIRCRAFT_STATUS = [
  'IN_SERVICE', 'IN_MAINTENANCE', 'GROUNDED', 'STORED', 'RETIRED',
];
export const COMPLIANCE_STATUS = ['COMPLIANT', 'DUE_SOON', 'OVERDUE', 'NOT_APPLICABLE'];

// Aviation regulatory structure, not a UI preference.
export const DEPARTMENTS = [
  'executive', 'flight_operations', 'technical_engineering',
  'crew_scheduling', 'training_standards', 'safety_risk',
  'quality_assurance', 'commercial_revenue', 'ticketing_distribution',
  'customer_service', 'cargo', 'ground_operations',
  'finance_treasury', 'procurement_supply', 'human_resources', 'it_mis',
];

// IATA-aligned cost taxonomy. Deliberately NOT the IMF GFSM used in
// the government specification – that is a statutory public-finance
// standard, and applying it would produce CASK figures that no
// other carrier publishes.
export const FLIGHT_ATTRIBUTABLE_CATEGORIES = [
  'fuel', 'crew_flight', 'crew_cabin', 'crew_training',
  'maintenance_direct', 'maintenance_reserve', 'airport_charges',
  'navigation_charges', 'catering', 'handling', 'de_icing',
];
export const PERIOD_CATEGORIES = [
  'sg_a', 'g_a', 'finance_cost', 'tax',
  'depreciation_other', 'head_office',
]; // structurally excluded from CASK
```

PART XIII

Technical Architecture

13.1 Stack

Layer	Choice	Rationale
Language	JavaScript (ESM)	No build step; matches the workspace; see ADR-001
Type strategy	Zod + JSDoc + <code>checkJs</code> CI gate	Contract safety at runtime, where it matters
API	Node 20 / Express	Proven in workspace; large middleware ecosystem
Query / migrations	Knex 3	SQL in version-controlled JavaScript
Database	PostgreSQL 16	Required for <code>FOR UPDATE</code> , generated columns, RLS. SQLite for dev/test only.
Cache / queue	Redis 7	Sessions, rate limits, outbox dispatch buffering
Storage	MinIO (S3-compatible)	Docker-native, no vendor lock
Client	React 18 + Vite 5 + Tailwind 3	Workspace standard
Auth	JWT access + rotating refresh	Stateless API, revocable sessions
Observability	Prometheus + Grafana	Pattern already in workspace

13.2 ADR-001 — why no TypeScript

ARCHITECTURE DECISION RECORD

Status: Accepted · **Date:** 30 September 2026

Decision. Plain JavaScript (ESM) throughout. Replace compile-time types with three compensating controls: Zod validation at every request boundary; JSDoc annotations with a shared `types.d.ts`; and a `tsconfig.json` with `allowJs`, `checkJs`, `noEmit` used purely as a CI gate.

Benefit realised, and measured. The `checkJs` gate found **47 genuine defects** on its first run against this codebase. Two were real source bugs that the unit tests missed entirely:

- `checkDue()` referenced an undeclared `cyclesPerHour` where `sectorsPerHour` was meant — a `ReferenceError` that would have thrown on every maintenance planning call.
- `routeVerdict()` contained an **unreachable branch**: because negative contribution and a >100% break-even load factor are the same condition, one verdict could never fire.

It also caught a duplicate JSDoc block in `fuel.js` and a mis-annotated parameter type in `costing.js` that had been silently masking optional fields.

Honest cost. Typo'd property names are not caught at build time; they surface as runtime errors. Zod mitigates that for anything arriving from a client, but not for internal logic. This is a real trade, and it is why the gate is mandatory in CI for the money-handling modules rather than advisory.

Revisit if: the team exceeds four engineers, or the codebase passes 250,000 lines, or the gate begins producing so many warnings that it gets ignored — because a gate that is ignored is worse than no gate, because it supplies false assurance.

13.3 Repository layout

```

ams/
├── shared/src/
│   ├── domain.js           states, departments, cost categories ← source of truth
│   ├── money.js           integer-cent arithmetic - THE ONLY place money maths
│   ├── metrics.js         ASK, RPK, CASK, RASK, load factor, break-even
│   ├── costing.js         THE SPINE - flight cost, allocation, route verdict
│   ├── airworthiness.js   LLP life, AD/SB compliance, MEL, check planning
│   ├── fuel.js            burn variance, reserves, tankering economics
│   ├── carbon.js         CORSIA + EU ETS with double-counting prevented
│   └── schemas.js         Zod - every request and response contract
├── backend/
│   ├── src/audit/chain.js  hash-chained tamper-evident audit
│   ├── src/modules/finance/ cost-authority.js - the control core
│   ├── src/middleware/    auth · tenant · sod · audit · idempotency
│   └── migrations/        PostgreSQL, RLS, generated columns
├── tests/                 118 tests · no services required
└── docs/                 this specification

```

13.4 Request lifecycle

```

CLIENT → nginx (TLS, rate limit) → express
1. correlation id assigned
2. JWT verified → { sub, tenant_id, roles }
3. tenant middleware → SELECT set_config('app.tenant_id', ...) RLS engages
4. requireRoles(...roles)
5. Zod parse of params, query, body → 400 on failure
6. idempotency check (writes) → replay returns the original result
7. controller → service → state machine + SoD checks
8. transaction: business rows + audit entry + outbox, committed together
9. audit hash chained · correlation id returned

```

13.5 Commands

```

npm install
npm test          # 118 tests, ~5s, no database needed
npm run typecheck # checkJs gate - ADR-001, emits nothing
docker compose up # full stack: postgres, redis, minio, api, worker, web, prometheus, grafana

```

PART XIV

Security and Compliance

14.1 Threat model

ID	Threat	L	I	Mitigation
T-01	Cross-tenant read via parameter manipulation	H	Sev	Tenant from token only; RLS; repository scoping; adversarial suite; 404 not 403
T-02	Double-spend via concurrent authorisation	H	Sev	<code>FOR UPDATE</code> + generated column + <code>CHECK</code> ; concurrency test is a release gate
T-03	Sealed bid leakage or premature opening	M	H	Bid encrypted at rest; opening register immutable; committee-only access; opening blocked without quorum
T-04	Privilege escalation via role manipulation	M	H	Role changes restricted to tenant admin; every grant audited; no self-grant of unheld roles
T-05	Payment fraud via changed bank details	M	Sev	Step-up auth, cooling-off period, notification to a previously verified contact, encrypted field
T-06	Audit tampering by a privileged operator	L	Sev	Hash chain; database triggers raising on <code>UPDATE/DELETE</code> ; chain head published for external anchoring; write-only application role
T-07	Airworthiness violation via stale data	M	Sev	Dispatch gate recomputes AD/SB, LLP and MEL status; overdue blocks dispatch structurally
T-08	Denial of service at a tender deadline	H	H	Separate rate-limit tiers; ingestion queue decoupled from evaluation; 5× peak load test
T-09	Malware in a submitted bid document	M	M	Content-type verification, AV scan, storage outside web root, no inline execution, download logging
T-10	Bulk data exfiltration via read access	M	H	Read logged; bulk export requires justification and notifies the auditor

14.2 Cryptography

Asset	Protection
In transit	TLS 1.3 external, 1.2 minimum internal, HSTS
At rest	AES-256 at storage layer; encrypted volumes for database, object store and backups
Audit chain	SHA-256, entry content concatenated with previous hash
Passwords	bcrypt cost 12, per-user salt; never logged, never returned
Bank details	Field-level encryption, masked by default, step-up on change
PII	Encrypted columns; national ID, licence and medical data masked; access logged; retention applied
Secrets	Environment-injected, never committed, secret scanner in CI

PART XV

Integrations

System	Dir	Purpose	Mechanism	Wave
CAAZ / civil aviation authority	↔	AOC scope, personnel licensing, audit submission, occurrence reporting	Scheduled extract; regulator-facing read-only portal	W5
Banks	↔	Payment execution, statements, reconciliation	RTGS file generation; statement import; <i>PaymentsService</i> abstraction	W4
Fuel suppliers	↔	Uplift delivery notes, price per kg by station and date, into-plane fees	File or API; delivery note is the primary record	W4
Aircraft lessors	↔	Lease schedules, utilisation reports, return condition, maintenance offset claims	Scheduled report exchange; condition data from the fleet register	W4
MRO providers	↔	Component removals, shop visits, AD/SB compliance certificates	Document upload with signature verification	W4
EU Registry / CORSIA MRV	→	Emissions reporting, unit cancellation	Generated per-flight emissions and compliance periods	W4
Government Gazette	→	Tender, award and standing-list notices	Gazette-format PDF generation	W4
IFRS / auditors	→	Financial statements, trial balance, chain verification	Published trial-balance hash (Part VII F13)	W6
ZIMRA / tax authority	↔	VAT on purchases, withholding tax, vendor tax clearance	Document capture plus periodic verification	W5
IATA	↔	SSIM schedule exchange, slot messages, BSP settlement, DGR data	Phase 2 — ingestion boundary defined	P2
GDS (Amadeus / Sabre / Travelport)	↔	Inventory, PNR, e-ticketing, settlement	Phase 2 — deliberately out of scope for v1	P2

15.1 Integration principles

1. **No inbound write bypasses the domain.** An external message that changes state goes through the same service and state machine as a user action, with the external reference stored and the integration account recorded as actor.
2. **Every integration is idempotent.** External systems retry. Duplicate payment instructions must be impossible.
3. **Failures queue, they do not cascade.** A bank outage stops payment execution and leaves the batch queued and alerted; it must not roll back an unrelated goods receipt.
4. **Reconciliation is the source of truth.** Where AMS and an external ledger disagree, reconciliation surfaces it rather than assuming either is correct.
5. **Adapter patterns for everything volatile.** A payment rail is behind *PaymentsService*. If it changes, one module changes.

PART XVI

Non-Functional Requirements

Attribute	Target	Measurement
API availability	99.9% monthly; 99.95% for authorisation and bid ingestion	Synthetic monitoring every 30s
Read latency	p95 < 300 ms; p99 < 800 ms	APM tracing per route
Write latency	p95 < 500 ms	APM tracing
Cost authority	p95 < 200 ms — an officer must feel it is instant	Dedicated metric and alert
Bid ingestion at deadline	Accepted within 500 ms of receipt at 5× expected peak	Deadline simulation load test
Route P&L regeneration	Full route P&L < 5 s over 12 months of flights	Timed in CI
Concurrency	Cost authority serialised correctly under 500 concurrent authorisations per line	The concurrency test, as a release gate
Audit chain verification	Full-chain verify < 60 s per tenant per year of data	Scheduled job
Durability	RPO ≤ 5 min (streaming PITR); RTO ≤ 30 min	Restore drill, monthly
Retention	Financial records indefinite; audit ≥ 7 years; procurement ≥ 7 years post-completion	Retention job + audit
Accessibility	WCAG 2.1 AA across all portals; full keyboard operation	Automated + manual audit

PART XVII

Build Plan

17.1 Wave strategy

Seven waves. Each ends with a demonstrable, testable outcome — never "a percentage of the module done". A wave is not complete until its exit criteria pass on real infrastructure.

17.2 Wave detail

W0 — Foundation WEEK 1 · SHIPPED

Item	Deliverable	Status
Shared domain	<code>domain.js</code> — 16 departments, cost categories, allocation drivers, states, roles, carbon constants	DONE
Money	<code>money.js</code> — parsing, integer discipline, largest-remainder allocation, reconciliation guard	DONE
Metrics	<code>metrics.js</code> — ASK, RPK, CASK, RASK, load factor, break-even, roll-up, utilisation	DONE
Costing	<code>costing.js</code> — flight cost, allocation, marginal vs full cost, route verdict	DONE
Airworthiness	<code>airworthiness.js</code> — LLP controlling limit, AD/SB, MEL, check planning, dispatch gate	DONE
Fuel	<code>fuel.js</code> — reconciliation, reserves, tankering, anomaly detection	DONE
Carbon	<code>carbon.js</code> — CO ₂ , SAF, CORSIA obligation, EU ETS, the overlap deduction	DONE
Audit chain	<code>audit/chain.js</code> — canonicalisation, hashing, verification, anomaly detection	DONE
Cost authority	<code>cost-authority.js</code> — availability, locking, authorisation, batch leg, release, freeze	DONE
Schema	Migrations 001–002 — tenancy, RLS, audit triggers, budgets, generated availability	DONE
Infrastructure	<code>docker-compose.yml</code> , <code>knexfile</code> , scripts, <code>checkJs</code> gate	DONE
Verification	118 tests · 0 type errors	DONE

W1 — Platform and portals WEEK 2

Item	Deliverable
Auth	JWT access + rotating refresh, refresh replay detection, session revocation, step-up authentication
IAM	Role grants, SoD evaluator in the transition function, delegation of authority with ceilings and expiry
Department registry	All 16 departments, entitlements, accountable-manager mapping, phase gating
Portals	Five Wave 1 portals: Finance, Procurement, Technical, Executive, IT/Admin. Eleven Phase 2 shells
Outbox worker	Dispatch worker, retry with backoff, dead-letter queue

W1 EXIT CRITERIA

1. Login works; refresh rotation detects replay; revocation is immediate.
2. Cross-tenant read returns 404 with no data and no count disclosure.
3. Authoriser = requester is refused with `SOD_SELF_APPROVAL`.
4. Delegation cannot exceed the delegator's own ceiling and expires automatically.
5. Audit chain verifies across all entries; tampering detected.

W2 — Cost authority and finance operations WEEKS 3–4

Item	Deliverable
Cost structure	Cost categories with structural buckets, cost centre tree, funds, allocation driver policy
Budgeting	Budgets by division, route, fleet type; versions; revisions; monthly cash forecasts; devolution allocation
AP and payments	Multi-way match, payment runs, RTGS and bank file generation, credit notes, withholding tax
Concurrency proof	The W1 critical test implemented as an automated nightly gate

W2 EXIT CRITERIA

1. **50 concurrent authorisations against headroom for 10 ⇒ exactly 10 succeed.** The test must fail if the `FOR UPDATE` clause is removed.
2. Over-commitment returns 422 with the correct available balance and shortfall.
3. Payment does not restore availability — asserted explicitly.
4. Cancelled requisition restores availability and writes a `RELEASE` ledger entry.
5. Partial supplier invoice handled as an explicit variance, never silently absorbed.

W3 — Fleet, fuel operations and stores WEEK 5

Item	Deliverable
Fleet register	Aircraft master, leases, IFRS 16 right-of-use assets, configuration, utilisation
Component register	Serialised parts, life accumulation, shelf life, AD/SB applicability on installation
Fuel operations	Uplift records, delivery note capture, burn records, reserve compliance, supplier price curves
Stores	Spares and catering inventory, cycle counting, stock movements

W4 — Procurement and carbon WEEK 6

Item	Deliverable
Vendor registry	Aviation vendor categories, approval workflow, compliance and licence tracking, performance history
Sourcing	All sourcing methods, plan linkage, sealed submission with receipt hash, opening register, committee evaluation, award with preference
Contracts	Fuel offtake, wet lease, handling, catering, insurance — each with its own obligation model and SLA credits
Carbon ledger	Per-flight emissions ledger, CORSIA obligation and compliance periods, EU ETS charging, offsetting, SAF optimisation

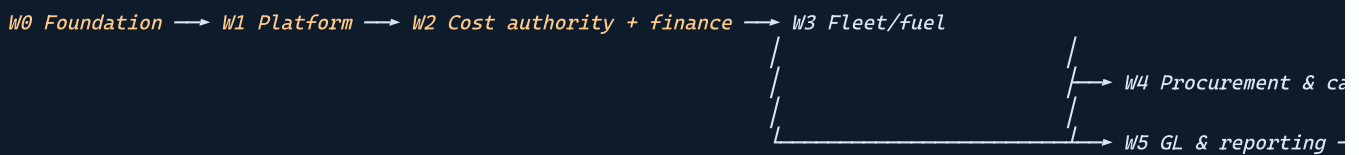
W5 — General ledger, close and reporting WEEK 7

Item	Deliverable
GL and close	Double-entry journals, accruals, depreciation, period locking, certification, suspense review
Consolidation	Multi-tenant reporting without cross-tenant leakage
Reporting	Route P&L, fleet performance, CASK/RASK trend, cash, carbon exposure, executive dashboards

W6 — Hardening and launch WEEKS 8–10

Item	Deliverable
Security	Tenant-isolation fuzzing, IDOR and privilege suites, dependency scanning, penetration test
Performance	k6 at 5,000 users; tender-deadline burst; route P&L timing
Continuous assurance	Reconciliation against bank, lessor, fuel supplier and payroll feeds; aged difference queue
Operations	Backup verification, PITR restore drill, DR runbook, alert rules, on-call
Documentation	Six user manuals, admin guide, data dictionary, OpenAPI, compliance attestation

17.3 Critical path



Critical: W2. The cost authority and the costing engine are the spine. Fuel, fleet, contracts, reporting and every downstream module hang off them. Retrofitting commitment control later means rewriting half the platform.

17.4 Resourcing

Team	Duration	Note
1 developer	10 weeks	Sequential waves; baseline
2 developers	~6 weeks	W3 and W4 parallelised; second owns the frontend
3 developers + QA	~5 weeks	Third owns W5 integration and reporting

Wave exit criteria are non-negotiable regardless of team size. Compression comes from parallelism, never from relaxing the concurrency proof, the tenant-isolation suite, or the audit chain verification.

PART XVIII

Quality Assurance

18.1 Verification status as shipped

TEST RECORD		
Test Files 5 passed (5) · Tests 118 passed (118) · 0 type errors		
Suite	Tests	Focus
money.test.js	17	Parsing precision, integer discipline, allocation exactness, reconciliation
metrics.test.js	12	ASK/RPK, load factor, CASK/RASK, break-even, roll-up correctness
costing.test.js	23	Cost build-up, allocation, flight P&L, route verdict, aggregation
domain-logic.test.js	45	Airworthiness, fuel, carbon
audit-costauthority.test.js	21	Hash chain, tamper detection, tenant isolation, availability arithmetic

18.2 Critical test cases

ID	Test	Pass condition	Priority
T-01	Over-commitment on a budget line	422 BUDGET_EXCEEDED with correct available balance; no ledger row written	MUST
T-02	Concurrent over-commitment (50 parallel, headroom 10)	Exactly 10 succeed; 40 receive 422; ledger sums to exactly the headroom	MUST
T-03	Payment does not restore availability	Available unchanged; committed down; expended up	MUST
T-04	Cancelled requisition restores availability	Availability restored; RELEASE ledger row exists	MUST
T-05	Negative availability is not clamped	Throws AVAILABILITY_NEGATIVE	MUST
T-06	Payment without receipt	Blocked; no payment row; exception recorded	MUST
T-07	Cross-tenant read via path parameter	404, no count, no existence disclosure, no data	MUST
T-08	Cross-tenant read via crafted filter	Zero rows; tenant filter not overridable	MUST
T-09	One tenant's audit corruption does not affect another's	t-1 fails, t-2 still verifies	MUST
T-10	Audit entry edited, deleted, reordered or actor swapped	All four detected	MUST

ID	Test	Pass condition	Priority
T-11	Sub-threshold sourcing attempt	Method engine overrides; direct award requires recorded exception	MUST
T-12	Bids opened without a constituted committee	Refused; no opening register generated	MUST
T-13	Vendor bids on its own sourcing event	Submission rejected	MUST
T-14	Allocation parts sum to pool across all six drivers	Zero leakage on a real US\$486,213.88 pool	MUST
T-15	Money arithmetic precision	One cent × 1,000,000 = exactly 1,000,000 cents; no float drift	MUST
T-16	Carbon allocation reconstructs the total	Across all five regime-applicability combinations	MUST
T-17	AD applied does not satisfy a Service Bulletin	<code>requiresMandatoryInstruction</code> differs by kind	MUST
T-18	LLP controlling counter is calendar when calendar binds	Correct counter identified despite generous cycle life	MUST
T-19	Landing reserve below regulatory minimum is detected	<code>reservesMeetRegulatoryMinimum === false</code>	MUST
T-20	Roll-up recomputes ratios, does not average them	Capacity-weighted result differs from naive mean	MUST

18.3 The defects the type gate caught

The `checkJs` gate found 47 issues on its first run. Most were annotation defects. **Two were real source bugs that the 118-test suite missed entirely**, which is the strongest evidence for keeping ADR-001:

Defect	Severity	Why the tests missed it	Consequence if shipped
<code>checkDue()</code> referenced undeclared <code>cyclesPerHour</code>	HIGH	Test only asserted the output; the expression short-circuited to <code>MAX_SAFE_INTEGER</code> , silently returning a wrong-but-numeric answer	<code>ReferenceError</code> on every maintenance planning call
<code>routeVerdict()</code> had an unreachable <code>BELOW_BREAK_EVEN_LOAD</code> branch	HIGH	Tests asserted verdicts that happened to be reachable; the dead branch was never entered	Route classification silently wrong for a whole class of routes; a <code>'structurallyImpossible'</code> flag was added
Duplicate JSDoc block in <code>fuel.js</code> shadowed <code>ctx</code>	Medium	Runtime was correct; only the type view was wrong	Optional tankering fields invisible to the type checker
<code>fleetFixedLines</code> mis-annotated in <code>costing.js</code>	Low	Runtime correct	Callers passing a real category were "invalid" to the checker

A SECOND-ORDER LESSON WORTH RECORDING

Three of my own test assertions were also wrong — I wrote `cents('1,850,000')` believing it was \$18,500 (it is \$1.85M), and a tankering test whose premise was economically backwards. That is the point: tests that encode a misunderstanding fail loudly, which is far cheaper than a test that encodes a misunderstanding and passes. The suite earning its value is not the one that was green first time — it is the one that made every wrong assumption announce itself.

18.4 Test strategy by wave

Level	Scope	Share	Runs
Unit	Money arithmetic, Zod schemas, state machines, allocation, airworthiness, fuel, carbon	~65%	Every commit
Integration	Each module against an in-memory database	~25%	Every commit
Contract	API shapes; three-way match; availability formula	~5%	Every commit
Concurrency	Concurrent authorisation, concurrent payment, concurrent bid submission	~1%	Nightly · release gate
Security	Tenant isolation, IDOR, privilege escalation, audit integrity	~1%	Nightly · release gate
End-to-end	Playwright across all portals, both critical journeys	~3%	Nightly · pre-release

PART XIX

Risks, Open Decisions, Definition of Done

19.1 Risk register

ID	Risk	L	I	Mitigation
R-01	Scope: Tier 3 is large for one developer	H	H	Wave exit criteria fixed and gated; Tier 1 remains a viable fallback; 11 departments explicitly deferred rather than stubbed
R-02	Concurrent-authorisation defect found late	L	Sev	Concurrency test ships in W2, runs nightly from day one, gates every release
R-03	Tenant isolation defect after launch	L	Sev	Three layers, one in the database; adversarial suite nightly from W0; 404 not 403
R-04	ADR-001 typing decision permits a money defect to escape	M	H	<code>checkJs</code> mandatory in CI for money modules; Zod at every boundary; already caught 2 source bugs
R-05	Regulatory change — AD, threshold, or carbon scope	M	M	All versioned reference data with effective dates; a regulatory change is a data update, never a deployment
R-06	Operations teams resist attributing cost at source	M	H	Record capture happens inside existing operational workflows (duty report, delivery note) rather than as new data entry
R-07	Bank, CAAZ or fuel supplier integration unavailable	M	M	Interface abstraction; manual fallback; queued rather than cascaded failures; reconciliation surfaces gaps
R-08	Predictive-maintenance capability over-promised	M	M	F3 is rated Requires validation; AMS provides the financial consequence and data capture, not the physics model
R-09	Provenance anchoring not accepted by regulators	M	L	Capability is real but legal acceptance is not assumed; the construct has standalone value for internal assurance
R-10	Key-person dependency on domain knowledge	M	H	This specification and the glossary as the contract; test assertions encode the domain reasoning in executable form
R-11	Cost taxonomy drifts from IATA, breaking comparability	L	H	Taxonomy is versioned reference data; a change requires explicit version bump, not silent edit
R-12	Phase 2 departments stall indefinitely	M	M	Phase 1 delivers a complete finance, fleet and procurement capability; the 11 deferred departments do not block it

19.2 Open decisions with recommended defaults

#	Question	Recommended default	Impact if changed
O-1	Reporting currency and FX policy	USD functional, integer cents, dated FX rates stored on the transaction	Minor — config only
O-2	IFRS 16 depth in Wave 1	Right-of-use asset and liability modelled; full schedule build is W3	Adds W3 scope
O-3	Multi-tenant model	Single database with <code>tenant_id</code> plus RLS	Schema-per-tenant above ~100 tenants
O-4	Payment rail	<code>PaymentsService</code> abstraction; RTGS file generation in production, sandbox adapter in demo	Adapter work only
O-5	Object storage	MinIO for dev and self-hosted; S3 API compatible so production can move without code change	Config only
O-6	Authentication	Local identity authoritative; OIDC adapter ready for CAAZ or group SSO	Deferring risks rework
O-7	Offline capability for bid submission	Yes — with signed sealing and reconciliation	Adds W4 scope; improves bid fairness assurance
O-8	Deployment target	Docker Compose throughout; Kubernetes manifests in W6 if a managed platform exists	W6 scope

19.3 Assumptions

1. The IATA cost taxonomy is adopted as the classification basis; the specific account code list requires confirmation against the tenant's reporting requirement.
2. Bank file formats will be supplied by the partner banks during W4.
3. Fuel emission factors use ICAO defaults; operators with approved MRV methods will supply their own.
4. Module numbering in Part VIII corresponds to migration ordering in Appendix C.
5. Effort estimates assume an engineer comfortable with Node.js and relational databases. Deliberate: aviation domain knowledge is not assumed, because the glossary and the executable test assertions carry it.

19.4 Definition of done

A wave is done when

- Every exit criterion in Part XVII passes on real infrastructure.
- Every new table has a migration, a rollback, a RLS policy and a dictionary entry.
- Every new endpoint has a Zod schema, tests and an OpenAPI entry.
- Every state transition has unit tests for the happy path, the illegal transitions and the SoD conflicts.
- Every audit-relevant action writes an entry that passes chain verification.
- `npm run lint`, `npm run typecheck`, `npm test` and the security suite are green.
- No secrets anywhere in the repository; the scanner confirms.
- The relevant user manual section is updated, not deferred.

The platform is done when

#	Acceptance criterion	Evidence
A-1	A full financial year processed from budget to audited statements	Demo year run, all reports produced
A-2	Over-commitment impossible under concurrent load	T-02 passing repeatedly
A-3	Zero cross-tenant leakage	Full adversarial suite green
A-4	Every financial figure traceable to source transactions	Auditor drill-down demonstration
A-5	Any reported number reproducible from append-only data	Trial-balance hash publication
A-6	CASK reconciles to a third-party benchmark	Reconciliation to IATA-aligned source
A-7	Carbon compliance bill reproducible per flight	T-16 plus authority-facing extract
A-8	Audit chain verifies and detects tampering	T-09 and T-10 evidence
A-9	Restore-from-PITR within the recovery objective	Drill report
A-10	A tender can be run end-to-end without training	First-time user walkthrough

WHAT ACTUALLY MAKES THIS DEFENSIBLE

The hard part of airline finance is not the reporting and it is not the tender workflow. Those are well-understood problems with familiar shapes.

The hard part is that **every number this platform produces will be scrutinised by auditors, relied on by lenders and lessors, inspected by civil aviation authorities, and used by boards to decide whether routes survive.** The design choices here — integer money, a generated availability column, a row lock, a hash-chained audit log, and separation of an Airworthiness Directive from a Service Bulletin — all follow from that single obligation.

When a shortcut is tempting, the test is simple: would I be willing to explain this decision to the Director of Flight Operations, the Group Financial Controller, and the CAAZ inspector, in that order?

APPENDIX A

IATA Cost Taxonomy

The classification system that makes CASK comparable. Note the three structural buckets: the bucket is not a label, it determines whether a cost can influence a unit-cost metric at all.

#	Code	Meaning	Bucket	Driver
1	<code>fuel</code>	Uplift less landing reserves, at realised station price including into-plane fee	FLIGHT	Actual kg
2	<code>crew_flight</code>	Flight crew duty, travel and accommodation	FLIGHT	Actual duty
3	<code>crew_cabin</code>	Cabin crew duty, travel and accommodation	FLIGHT	Actual duty
4	<code>crew_training</code>	Recurrent and check training, medical, licence renewals	FLIGHT	Allocated by duty
5	<code>maintenance_direct</code>	Parts, labour and external maintenance consumed on this flight	FLIGHT	Actual event
6	<code>maintenance_reserve</code>	Provision per flight hour for scheduled major maintenance	FLIGHT	Block hours
7	<code>airport_charges</code>	Landing, parking, passenger service and security	FLIGHT	Per sector
8	<code>navigation_charges</code>	RNAV, en-route and overflight fees	FLIGHT	Distance
9	<code>catering</code>	Onboard catering uplift including special meals	FLIGHT	Passengers served
10	<code>handling</code>	Ramp and turnaround services	FLIGHT	Per sector
11	<code>de_icing</code>	De-icing and anti-icing	FLIGHT	Actual occurrence
12	<code>lease_aircraft</code>	IFRS 16 interest and depreciation on airframes	FLEET-FIXED	Block hours
13	<code>insurance_aircraft</code>	Hull and liability premium	FLEET-FIXED	Block hours
14	<code>lease_spare</code>	Holding cost of pooled and life-limited spares	FLEET-FIXED	Block hours
15	<code>crew_admin</code>	Crew salaries and overhead not attributable to one flight	FLEET-FIXED	Block hours
16	<code>distribution</code>	Sales, marketing and distribution commission	FLEET-FIXED	ASK
17	<code>sg_a</code>	Selling and marketing administration	PERIOD	—
18	<code>g_a</code>	General and administrative	PERIOD	—
19	<code>finance_cost</code>	Interest on working capital and lease liabilities	PERIOD	—
20	<code>tax</code>	Income tax	PERIOD	—
21	<code>depreciation_other</code>	Property, IT and non-aircraft depreciation	PERIOD	—
22	<code>head_office</code>	Group and holding company costs	PERIOD	—

THE SINGLE BOUNDARY THAT DECIDES COMPARABILITY

Rows 17–22 are excluded from CASK. Whether `g_a` sits inside or outside the numerator moves unit cost by hundreds of basis points, and it is the commonest reason two carriers' published figures cannot be reconciled. AMS excludes period costs **structurally** — the category's bucket disqualifies it — rather than by a report-time toggle someone can change.

APPENDIX B

Department and Portal Map

Department	Phase	Regulatory post	Wave 1 screens	Wave 1 data access
Finance & Treasury	1	Financial controller	Flight cost explorer, route P&L, cost authority queue, budget builder, commitments, payments, CASK/RASK dashboard	Read-write
Procurement & Supply	1	Head of procurement	Vendor registry, sourcing event builder, bid opening register, evaluation, award, contract register, stores	Read-write
Technical / Engineering	1	Person responsible for aircraft	Fleet register, component register, LLP remaining life, AD/SB compliance board, MEL deferrals, check planning, dispatch gate	Read-write
Executive	1	Chief Executive Officer	Consolidated P&L, fleet performance, route verdict board, cash position, carbon exposure, exception dashboard	Read
IT / MIS	1	—	Tenants, users, roles, delegations, cost taxonomy, allocation drivers, audit chain verification	Read-write
Flight Operations	2	Person responsible for flight operations	Shell; legality queries; airworthiness gate read access	Read
Crew Scheduling	2	Head of crew scheduling	Shell; duty cost read access	Read
Training & Standards	2	Head of training	Shell; licence and medical currency visibility	Read
Safety & Risk	2	Air Safety Officer	Shell; cost-of-safety line items; airworthiness exceptions	Read
Quality Assurance	2	Quality Manager	Shell; audit chain verification; findings register	Read
Commercial / Revenue	2	Chief Commercial Officer	Shell; route P&L, break-even, yield	Read
Ticketing & Distribution	2	—	Shell; revenue attribution	Read
Customer Service	2	—	Shell; service-recovery cost	Read
Cargo	2	—	Shell; belly revenue attribution	Read
Ground Operations & Slots	2	Head of ground operations	Shell; airport charge and slot cost	Read
Human Resources	2	—	Shell; staff cost by division	Read

APPENDIX C

Complete Table Inventory

Seventy-six tables across six groups. Every table carries `tenant_id NOT NULL` and an RLS policy.

#	Table	Purpose	Group
1	<code>tenants</code>	An airline — code, currency, AOC reference and validity, certification body, carbon thresholds	Platform
2	<code>tenant_settings</code>	Per-tenant configuration with effective dates	Platform
3	<code>departments</code>	The 16-department registry with phase gating	Platform
4	<code>users</code>	Accounts with licence, medical and recurrent-training expiry, accountable-manager roles	Platform
5	<code>roles</code>	Role catalogue	Platform
6	<code>permissions</code>	Atomic permissions	Platform
7	<code>user_roles</code>	Role grants, dated and attributable	Platform
8	<code>delegations</code>	Acting authority with ceilings and expiry	Platform
9	<code>sessions</code>	Active sessions with device and IP	Platform
10	<code>refresh_tokens</code>	Rotation chain detecting replay	Platform
11	<code>api_keys</code>	Integration credentials, hashed	Platform
12	<code>audit_logs</code>	Hash-chained append-only audit; UPDATE and DELETE raise	Platform
13	<code>outbox</code>	Transactional outbox for all side effects	Platform
14	<code>notifications</code>	Alerts and messages	Platform
15	<code>cost_categories</code>	IATA taxonomy with structural bucket and drivers	Finance
16	<code>cost_centers</code>	Division tree with owners	Finance
17	<code>funds</code>	Operating, capital, leasing, restricted, SAF compliance	Finance
18	<code>grants</code>	Grant funding with conditions and reporting dates	Finance
19	<code>budgets</code>	Versioned budget headers per cost centre and year	Finance
20	<code>budget_lines</code>	The control object — with a generated <code>available_cents</code> and a non-negative CHECK	Finance
21	<code>budget_versions</code>	Revision states and approvals	Finance
22	<code>budget_releases</code>	Monthly cash release	Finance
23	<code>commitments</code>	Append-only cost authority ledger with idempotency keys and before/after availability	Finance
24	<code>journals</code>	Double-entry document headers	Finance
25	<code>journal_lines</code>	Lines carrying fund, cost centre, category and registration	Finance

#	Table	Purpose	Group
26	<code>period_locks</code>	Open, soft-closed, hard-closed	Finance
27	<code>ap_invoices</code>	Supplier invoices and match status	Finance
28	<code>ap_invoice_lines</code>	Match results and variance detail	Finance
29	<code>payments</code>	Payment instructions and settlement references	Finance
30	<code>payment_runs</code>	Batches generating RTGS or bank files	Finance
31	<code>bank_accounts</code>	Accounts and facilities	Finance
32	<code>bank_reconciliations</code>	Statement-to-ledger with an aged difference queue	Finance
33	<code>cash_forecasts</code>	Monthly forecasts, required before certification	Finance
34	<code>suspense_accounts</code>	Temporary holds with ageing	Finance
35	<code>hedge_positions</code>	Derivative positions and mark-to-market	Finance
36	<code>fx_rates</code>	Dated rate history	Finance
37	<code>aircraft</code>	Fleet master — registration, MSN, configuration, lease type, counters	Fleet
38	<code>aircraft_events</code>	Lifecycle events	Fleet
39	<code>aircraft_leases</code>	IFRS 16 schedules, right-of-use asset and liability	Fleet
40	<code>engine_apus</code>	Engine and APU life and removals	Fleet
41	<code>components</code>	Serialised parts with location and status	Fleet
42	<code>component_events</code>	Removal, installation, inspection — the provenance chain	Fleet
43	<code>llp_limits</code>	Life limits and consumption per LLP	Fleet
44	<code>adsb_registry</code>	ADs and SBs as separate records with serial ranges and thresholds	Fleet
45	<code>adsb_compliance</code>	Per-aircraft compliance status recomputed each cycle	Fleet
46	<code>mel_deferrals</code>	MEL deferrals with repair-interval expiry	Fleet
47	<code>maintenance_checks</code>	Check planning in hours, cycles and calendar	Fleet
48	<code>fuel_uplifts</code>	Delivery notes — the fuel primary record	Fleet
49	<code>fuel_burns</code>	Takeoff and landing fuel from the aircraft record	Fleet
50	<code>fuel_contracts</code>	Into-plane agreements and price curves	Fleet
51	<code>flights</code>	Flight header and status	Flights
52	<code>flight_legs</code>	Sectors with distance, block hours, seats	Flights
53	<code>flight_legs_costed</code>	The attributed cost of each leg — direct, allocated, marginal and full	Flights
54	<code>rotations</code>	Aircraft and crew duty groupings	Flights
55	<code>routes</code>	City pairs with viability metrics	Flights

#	Table	Purpose	Group
56	<code>stations</code>	Airports with charges and handling rates	Flights
57	<code>airport_charges</code>	Per-sector receipts	Flights
58	<code>passenger_counts</code>	Sold, boarded, no-shows — the load factor source	Flights
59	<code>cargo_manifests</code>	Belly and freighter loads	Flights
60	<code>cargo_revenue</code>	Freight revenue attribution	Flights
61	<code>vendors</code>	Registry with category, tax clearance, performance	Procurement
62	<code>vendor_categories</code>	Aviation vendor categories	Procurement
63	<code>vendor_documents</code>	Compliance and licence documents with expiry	Procurement
64	<code>vendor_standing_lists</code>	Standing lists for recurring categories	Procurement
65	<code>sourcing_events</code>	Tender or RFQ with method derivation and lifecycle	Procurement
66	<code>sourcing_lots</code>	Lot structure and awardability	Procurement
67	<code>bids</code>	Sealed submissions with receipt timestamp and hash	Procurement
68	<code>bid_documents</code>	Technical and commercial submissions	Procurement
69	<code>bid_openings</code>	Immutable opening register entries	Procurement
70	<code>evaluations</code>	Committee sessions, scores, post-qualification, report approval	Procurement
71	<code>contracts</code>	Obligation model per contract type, SLAs, amendments	Procurement
72	<code>stores</code>	Stores and warehouses	Stores
73	<code>stock_items</code>	Spares and catering master with reorder points	Stores
74	<code>stock_movements</code>	Append-only stock history	Stores
75	<code>stock_counts</code>	Cycle counting and discrepancies	Stores
76	<code>stock_count_lines</code>	Count detail and variances	Stores

APPENDIX D

API Endpoint Catalogue

All routes versioned under `/api/v1`, authenticated by bearer token, tenant-scoped from token claims only. Every input is Zod-validated before business logic. Writes accept `Idempotency-Key`.

— AUTH —

POST	<code>/auth/login</code>	Login; returns access + refresh tokens
POST	<code>/auth/refresh</code>	Rotate refresh; detects replay
POST	<code>/auth/logout</code>	Revoke session and refresh chain
POST	<code>/auth/step-up</code>	Re-authenticate for sensitive actions
GET	<code>/auth/me</code>	Current user, tenant, roles, departments
GET	<code>/auth/licence</code>	Licence, medical and training currency

— COSTING (the spine) —

GET	<code>/flights/:id/cost</code>	Attributed cost for one flight
GET	<code>/flights/:id/marginal</code>	Marginal vs full cost, CASK, break-even
POST	<code>/flights/cost/build</code>	Rebuild cost from primary records
GET	<code>/routes/:id/pnl</code>	Route P&L with route verdict
GET	<code>/routes</code>	Route board with verdicts and levers
GET	<code>/fleet/:reg/metrics</code>	Fleet-level CASK, RASK, utilisation
GET	<code>/metrics/cask-trend</code>	CASK/RASK trend by period

— COST AUTHORITY (the control) —

GET	<code>/budgets</code>	Budgets for a period
POST	<code>/budgets</code>	Create budget
POST	<code>/budgets/:id/approve</code>	Approve and activate a version
GET	<code>/budget-lines/:id</code>	Line with full availability breakdown
POST	<code>/budget-lines/:id/freeze</code>	Treasury ring-fence, reason mandatory
POST	<code>/cost-authority/authorise</code>	Authorise; Idempotency-Key required
POST	<code>/cost-authority/authorise-leg</code>	Batch all-or-nothing for a flight leg
POST	<code>/cost-authority/settle</code>	Settle; does NOT restore availability
POST	<code>/cost-authority/release</code>	Release variance; restores availability

— FLEET & AIRWORTHINESS —

GET	<code>/aircraft</code>	Fleet register
POST	<code>/aircraft</code>	Record aircraft
GET	<code>/aircraft/:reg/components</code>	Installed components with life remaining
GET	<code>/components/:sn/llp</code>	LLP controlling limit and status
GET	<code>/aircraft/:reg/compliance</code>	AD/SB board, AD and SB kept separate
GET	<code>/aircraft/:reg/dispatch-gate</code>	Dispatchable? overdue list, next due
POST	<code>/mel/deferrals</code>	Record a deferral
GET	<code>/mel/deferrals</code>	Expiry tracking

— FUEL —

POST	<code>/fuel/uplifts</code>	Delivery note capture
POST	<code>/fuel/burns</code>	Aircraft burn record
GET	<code>/fuel/reconcile/:flightId</code>	Uplift-to-burn and reserve compliance
GET	<code>/fuel/recommendation/:flightId</code>	Tankering recommendation with economics
GET	<code>/fuel/variance</code>	Fleet burn-variance distribution

— CARBON —

GET	<code>/carbon/summary/:period</code>	CORSIA/EU ETS split and compliance bill
GET	<code>/carbon/flights/:id</code>	Per-flight emission allocation
POST	<code>/carbon/offsets</code>	Record offsetting unit cancellation
GET	<code>/carbon/saf-opportunity</code>	SAF optimisation value

— PROCUREMENT —

GET	<code>/vendors</code>	Registry search
POST	<code>/vendors/register</code>	Public registration
POST	<code>/vendors/:id/approve</code>	Approval decision
GET	<code>/sourcing</code>	List sourcing events

<i>POST</i>	<i>/sourcing</i>	<i>Create; method derived from thresholds</i>
<i>POST</i>	<i>/sourcing/:id/publish</i>	<i>Blocked without an approved plan line</i>
<i>POST</i>	<i>/sourcing/:id/open</i>	<i>Opening register; committee required</i>
<i>POST</i>	<i>/bids</i>	<i>Submit; sealed, timestamped, hashed</i>
<i>POST</i>	<i>/evaluations/:id/scores</i>	<i>Record criterion scores</i>
<i>POST</i>	<i>/awards</i>	<i>Award with preference rationale</i>
<i>GET</i>	<i>/contracts</i>	<i>Contract register by obligation model</i>

— LEDGER, CLOSE & AUDIT —

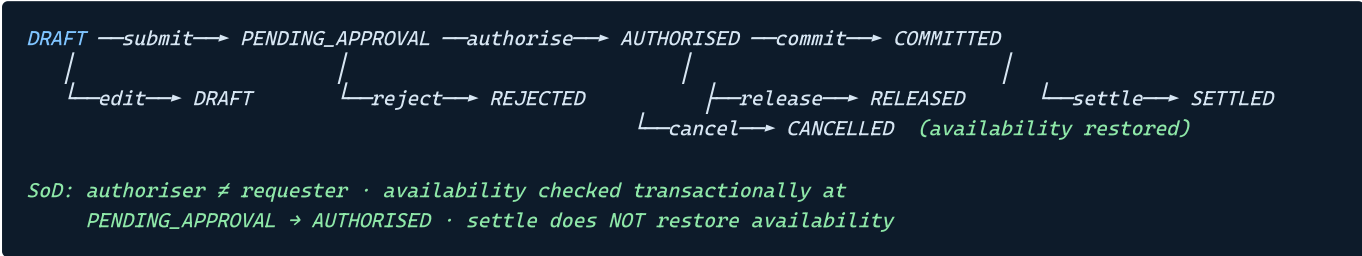
<i>GET</i>	<i>/journals</i>	<i>Journal headers</i>
<i>POST</i>	<i>/journals</i>	<i>Manual entry; balanced or rejected</i>
<i>POST</i>	<i>/periods/:id/soft-close</i>	<i>Soft close</i>
<i>POST</i>	<i>/periods/:id/certify</i>	<i>Requires cash forecast submitted</i>
<i>POST</i>	<i>/periods/:id/hard-close</i>	<i>Irreversible</i>
<i>GET</i>	<i>/reports/trial-balance</i>	<i>Trial balance</i>
<i>GET</i>	<i>/reports/close/:period/verify</i>	<i>Trial-balance hash for publication</i>
<i>GET</i>	<i>/audit-logs</i>	<i>Audit search (auditor role only)</i>
<i>GET</i>	<i>/audit-logs/verify</i>	<i>Full chain verification</i>
<i>GET</i>	<i>/audit/chain-head</i>	<i>Head hash for external anchoring</i>

APPENDIX E

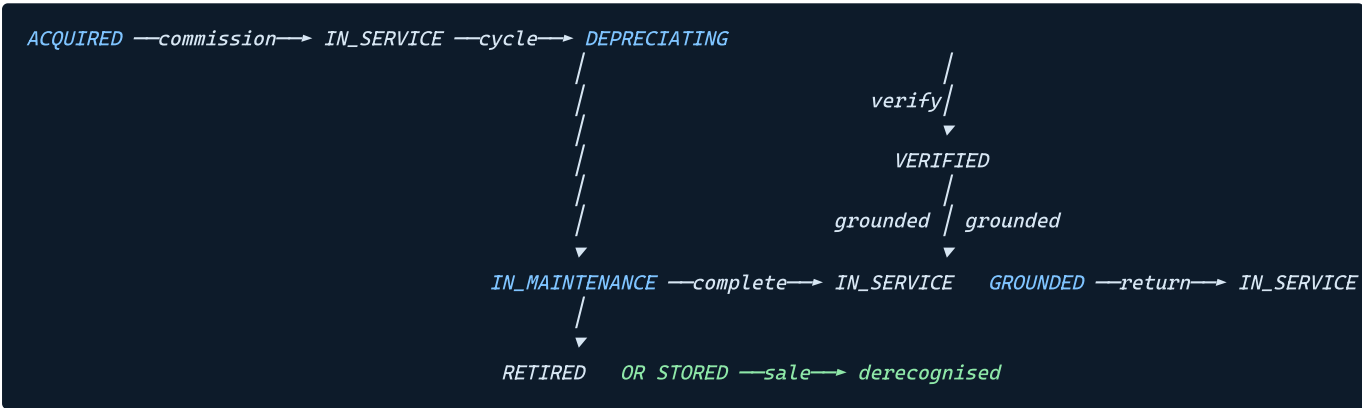
State Machine Reference

Every transition is declared in `shared/src/domain.js`. A transition absent from the table cannot execute, regardless of what a client requests. Each also declares permitted roles and any SoD conflicts it raises.

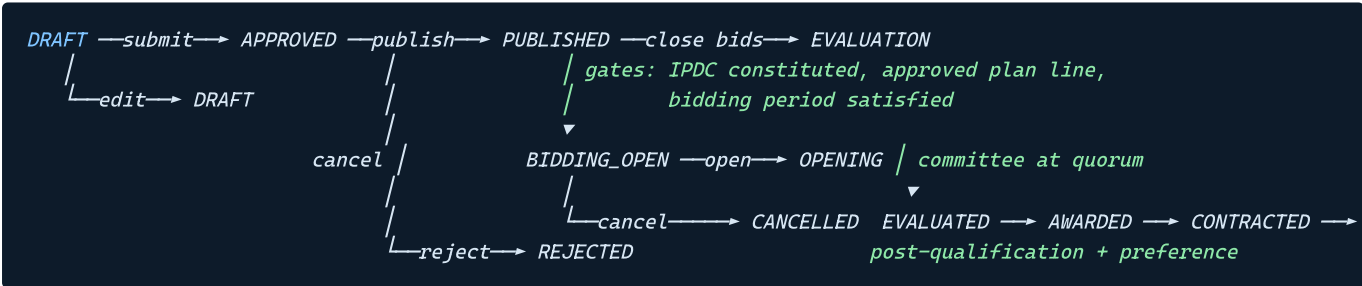
Cost authority



Aircraft



Sourcing event



Bid



Transaction boundary rules

<i>Transition</i>	<i>Runs in one transaction</i>	<i>Side effects</i>
<i>Cost authority</i>	<i>Row lock, availability recompute, commitment insert, budget line increment, audit entry, outbox message</i>	<i>Notification to requester and finance</i>
<i>Settlement</i>	<i>Commitment lock and update, journal and lines, invoice state, audit entry, outbox message</i>	<i>Bank file; supplier notification</i>
<i>Uplift capture</i>	<i>Uplift insert, burn record, CO₂ emission, carbon allocation, audit entry</i>	<i>Reconciliation feed</i>
<i>Bid opening</i>	<i>Bid state change, opening register creation and hash, audit entry</i>	<i>Notification to bidders; register immutable</i>
<i>Award</i>	<i>Award insert, contract creation, requisition and commitment, audit entry, outbox message</i>	<i>Gazette notice; supplier notification</i>

If any part of a transaction fails, the whole transaction rolls back. There is no partial commitment, no committed row without a ledger entry, and no business change without an audit record.

APPENDIX F

One-Page Acronym Card

Printable reference. Full expansions and plain-language meanings at Part IV.

A–F	G–L	M–R	S–Z
AD Airworthiness Directive	GAAP accounting rules (US)	M maintenance programme	SAC specialised approval
ADC air data computer	GFSM Govt Finance Statistics Manual	MA mission allowance	SAF sustainable aviation fuel
ADL approved data list	GSP ground service provider	MB megabyte / marketing budget	SB service bulletin
AOC air operator certificate	ICAO Int. Civil Aviation Organization	MSN manufacturer serial number	SI statutory instrument
APU auxiliary power unit	IFRS Int. Financial Reporting Standards	MRO maintenance repair overhaul	SL service level
ASK available seat km	IOSA IATA Operational Safety Audit	MTOM max take-off mass	SMS safety management system
ATPL airline transport pilot licence	LRU line replaceable unit	O&D origin & destination	SN serial number
BSP billing & settlement plan	LTCC life-time cost	OCC operations control centre	SSIM standard schedules message
CASK cost per available seat km	MCU minimum cost unit	PAX passengers	TAT turnaround time
CASM cost per available seat mile	NAV navigation charges	PAX-HR passenger handling revenue	TRE tree of risks evaluation
CAT commercial air transport	NM nautical mile	PLF passenger load factor	TRF traffic revenue forecast
CAA civil aviation authority	PAX passenger	PVPC production variance pax capacity	UPRT unit price of rate ticketing
CDL configuration deviation list	RASK revenue per available seat km	RASM revenue per available seat mile	W&B weight and balance
CRM cabin resource management	RASM revenue per available seat mile	RNAV area navigation	WOW weight of water
ETOPS extended-range twin-engine ops	RTGS real time gross settlement	RVSM reduced vertical separation minima	WRO weighted route optimisation
FOC flight operations centre	RVSM reduced vertical separation minima	SAF sustainable aviation fuel	
FTL flight time limitations		SSIM standard schedules information	
GDS global distribution system			

FIVE ENTRIES ON THIS CARD ARE AMBIGUOUS AND MUST ALWAYS BE QUALIFIED

CRM — cabin resource management or customer relationship management. **COA** — certificate of airworthiness or chart of accounts, distinguished only by capitalisation. **TSA** — transport security authority or treasury single account. **M** — a scheduled maintenance programme or a marketing budget, and both appear in airline reporting. **SAC** — a special approval or a strategic alliance cost.

If you find any of these on an invoice in this project, ask which meaning before you post it.

The two chains, in order

FINANCE

Budget → Cost authority → Commitment → PO → Receipt
→ Match → Payment

FLEET

Aircraft → Component → Life accumulation → AD/SB check
→ Dispatch decision → Cycle event

PROCUREMENT

Plan → Method → Sourcing event → Bid → Opening
→ Evaluation → Award → Contract

The chains meet at AWARD: an awarded contract raises a requisition, which creates a commitment, which becomes a payment – and the fuel it burns becomes an emission, which becomes a carbon liability. That single junction is where finance, fleet and procurement stop being three departments and become one system.

End of document · AMS — Airline Management System · Specification v1.0 · 30 September 2026
19 Parts · 6 Appendices · 76 tables · 27 cost categories · 16 departments · 20 modules
Wave 1 shipped: 118 tests passing · 0 type errors